

KPZ Universality in Combinatorial Systems

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Chapter 1

Preliminaries

1.1 Basic Definitions

Consider a measure space $(\Omega, \mathcal{M}, \mu)$.

Definition 1.1.1. We call Ω a **probability space** if $\mu \geq 0$ and $\mu(\Omega) = 1$. In this case, the measure μ is called a **probability measure** and the elements of the σ -algebra $\mathcal{M}$ are called **events**. Two events A and B are said to be **independent** if $\mu(A \cap B) = \mu(A)\mu(B)$.

We can model experiments on the sample space Ω by suitably nice functions on Ω . These are called random variables:

Definition 1.1.2. A **random variable** is a measurable function X from a probability space $(\Omega, \mathcal{M}, \mu)$ into a measurable space $(E, \mathcal{N}, \nu)$.

The requirement that a random variable X map into a measurable space and itself be a measurable function should be easily seen to be a natural one since it ensures that for any subset $F \subseteq E$ of the target space E , the set $X^{-1}(F) = \{\omega \in \Omega : X(\omega) \in F\}$ is measurable; or, in less mathematical terms, that for any collection of outcomes F , one can always ask the question “how likely is it that the outcome of the experiment X is among those in F ?” We denote this by $\mathbb{P}(X \in F) := \mu(X^{-1}(F))$.

To avoid confusion, it should be helpful to remark that it is common practice to implicitly define random variables by merely specifying the codomain space and prepending the word “random.” For instance, when one writes “let X be a random subset of $\mathbb{R}^2$ ”, the reader should interpret X as a random variable which takes values in the set of subsets of $\mathbb{R}^2$ and *not* a fixed subset of the plane chosen arbitrarily.

Definition 1.1.3. If $X: \Omega \rightarrow E$ is a random variable, we define the **expectation** $\mathbb{E}X$ of X to be the average value of X over all events in the probability space Ω , weighted according to the probability measure μ :

$$\mathbb{E}X := \int_{\Omega} X \, d\mu.$$

Note that because $\mathbb{E}X$ is defined by integrating the random variable X , it inherits all of the familiar properties of integrals. In particular,

Proposition 1.1.1. If $X, Y: \Omega \rightarrow \mathbb{R}$ are real-valued random variables and $X, Y \in L^1$, then:

(a) $\mathbb{E}(aX + Y) = a\mathbb{E}X + \mathbb{E}Y$ for all real numbers a ;

(b) If $X \geq Y$ then $\mathbb{E}X \geq \mathbb{E}Y$. □

Definition 1.1.4. If $X: \Omega \rightarrow E$ is a random variable, we define the **variance** $\text{Var}(X)$ of X to be the expected value of the squared deviation of X from its mean $\mu = \mathbb{E}X$:

$$\text{Var}(X) := \mathbb{E}[(X - \mu)^2].$$

1.2 Probability Distributions

When one has access to the datum of a random variable, they can ask questions out about the probabilities associated to subsets of the target space, i.e. outcomes. This induces a probability measure on the target space, called the “distribution” of the random variable:

Definition 1.2.1. Given a random variable $X: \Omega \rightarrow E$, the **probability distribution** or **law** of X is the pushforward $X_*\mu$ of the probability measure μ on Ω to a probability measure on E , defined via

$$X_*\mu(F) := \mu(X^{-1}(F)) = \mathbb{P}(X \in F).$$

Two random variables X, Y with the same target space are said to be **identically distributed** if they have identical probability distributions. We also say that X and Y are **independent** if receiving information about one of them does not change the probability distribution of the other; that is, if $\mathbb{P}(X \in A \text{ and } Y \in B) = \mathbb{P}(X \in A)\mathbb{P}(Y \in B)$ for all subsets A, B of the target space.

In certain cases, the probability distribution of a random variable X can be encoded by a function, called the “probability density function” of the random variable.

Definition 1.2.2. A random variable $X: (\Omega, \mathcal{M}, \mu) \rightarrow (E, \mathcal{N}, \nu)$ is said to have **density** f if for every subset $F \subseteq E$ we have that

$$\mathbb{P}(X \in F) = \int_F f \, d\nu.$$

Equivalently, X has density f if and only if f is the Radon-Nikodym derivative $dX_*\mu/d\nu$.

For instance, the normal distribution with mean μ and variance σ^2 , denoted $\mathcal{N}(\mu, \sigma^2)$, has density

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}.$$

If the random variable X is real-valued, then we can do even better and condense the the information encoded by the density function into a **cumulative distribution function** F by integrating:

$$F(x) := \int_{-\infty}^x f = \mathbb{P}(X \leq x).$$

1.3 Convergence of Random Variables

Given a sequence of random variables $X_1, X_2, \dots$ (all with a common domain and target space), there are several ways in which the X_n may converge to a limiting random variable X .

Definition 1.3.1. We say that the X_n converge **almost surely** (or **with probability 1**) to X , denoted $X_n \xrightarrow{a.s.} X$, if $\mathbb{P}(X_n \rightarrow X) = 1$, or in other words if the X_n converge to X pointwise almost everywhere on the event space Ω . Note that this definition only makes sense if the random variables X_n, X take values in a topological space.

Definition 1.3.2. We say that the X_n converge **in probability** to X , denoted $X_n \xrightarrow{p} X$ or $X_n \xrightarrow{\mathbb{P}} X$, if for any $\varepsilon > 0$ we have that $\mathbb{P}(|X_n - X| > \varepsilon) \rightarrow 0$. Note that this definition only makes sense if the random variables X_n, X take values in a normed space.

Definition 1.3.3. We say that the X_n converge **in distribution** to X , denoted $X_n \xrightarrow{d} X$, if the laws of the X_n converge weakly to the law of X ; that is, for all continuous bounded functions f , we have

$$\int f \, d\mathbb{P}_{X_n} \rightarrow \int f \, d\mathbb{P}_X.$$

1.4 Some Classical Theorems

Theorem 1.4.1 (Weak Law of Large Numbers). *Let $X_1, X_2, \dots$ be a sequence of i.i.d. random variables with expectation μ . Then*

$$\frac{S_n}{n} \xrightarrow{p} \mu \text{ as } n \rightarrow \infty$$

where $S_n = X_1 + \dots + X_n$.

Theorem 1.4.2 (Strong Law of Large Numbers). *Let $X_1, X_2, \dots$ be a sequence of i.i.d. random variables with expectation μ . Then*

$$\frac{S_n}{n} \xrightarrow{\text{a.s.}} \mu \text{ as } n \rightarrow \infty.$$

Theorem 1.4.3 (Central Limit Theorem). *Let $X_1, X_2, \dots$ be a sequence of i.i.d. random variables with expectation μ and variance $0 < \sigma^2 < \infty$. Then*

$$\frac{S_n}{\sqrt{n}} \xrightarrow{d} \mathcal{N}(\mu, \sigma^2) \text{ as } n \rightarrow \infty.$$

Chapter 2

The KPZ Universality Class

2.1 Last Passage Percolation (LPP) Models

In this section, we give an example of an exactly solved model of the KPZ universality class. This model is itself part of a more general class of models (all believed to be part of the KPZ class), known as *last passage percolation* models, or LPP models for short.

The idea behind this class of models is as follows. Consider the integer lattice $\mathbb{Z}^2$. If, to each member (i, j) of the lattice, we assign a random “weighting” $\omega_{(i,j)}$, then it is straightforward to see that we can assign a weighting to any *path* π in the lattice by simply summing the weights of the points which lie along π :

$$\omega_\pi := \sum_{(i,j) \in \pi} \omega_{(i,j)}.$$

A question one can then ask is the following: among all possible *up-right* paths proceeding from the origin to a fixed point (m, n) in the lattice, which has the highest weighting? If we let $\Pi_{(m,n)}$ denote the set of such paths, then because we restricted our attention to up-right paths the answer is easily seen to be

$$\tau_{(m,n)} := \max_{\pi \in \Pi_{(m,n)}} \omega_\pi = \max_{\pi \in \Pi_{(m,n)}} \sum_{(i,j) \in \pi} \omega_{(i,j)},$$

which we define to be the *last passage percolation time* of the point (m, n) . In general, we replace the lattice $\mathbb{Z}^2$ with a random collection of points prescribed by a *point process*:

Definition 2.1.1. A **polish space** is a complete, separable metric space. Given a polish space Λ , a **point process** on Λ is a random integer-valued Radon measure $\mathcal{X}$ on Λ . Such a process is said to be **simple** if $\mathcal{X}$ almost surely assigns value at most 1 to singletons.

Remark. Given a borel set U , one should think of the value as $\mathcal{X}(U)$ as the number of “points” inside U ; hence why we call such a measure a point process.

The overarching goal of all LPP models is to find an explicit formula which defines the cumulative distribution function of $\tau_{(m,n)}$ exactly. In this section, we shall see that this can be done in a few special cases; virtually all others remain open.

2.1.1 The Hammersley Problem

A useful motivator for the subsequent sections is the following last passage percolation problem. In the square with side length N and lower-left corner the origin $(0, 0)$, consider a Poisson point process of unit intensity (meaning a point process in which the Radon measure $\mathcal{X}$ is Poisson-distributed and the expected number of points in any given borel set U is equal to its area). The *Hammersley Problem* is to determine the maximum number of points that can be collected by travelling from $(0, 0)$ to (N, N) via an up-right path. It turns out that this problem is equivalent to finding the longest increasing subsequence in a random permutation, via the following simple observation. We can order the Poisson points in the square according to the order of their horizontal projections. If we then insert the coordinates $(x, y) \sim \binom{x}{y}$ of each such point into a 2-rowed matrix according to this ordering, then we’ll get something like

$$\begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 1 & 4 & 3 & 5 & 2 \end{pmatrix}$$

and the collection of Poisson points can then be identified with the permutation $\sigma: (1, 2, 3, 4, 5) \mapsto (1, 4, 3, 5, 2)$. The crucial aspect of this perspective is that the maximum number of such points which can be collected by an up-right path from $(0, 0)$ to (N, N) is equal to the length $L(\sigma)$ of the longest increasing subsequence of this permutation.

2.1.2 Patience Sorting and the RS(K) Correspondence

The previous example motivates the following question: given a permutation σ , how can we *compute* $L(\sigma)$? Of course, we can do this simply by brute force when the order n of σ is small, however it is not difficult to see that this approach quickly becomes infeasible as n grows.

A different approach which is both simple and computationally inexpensive is the **patience sorting** algorithm, which is defined as follows. Given a permutation σ , let us write $\sigma = (\sigma(1), \sigma(2), \dots, \sigma(n))$. Beginning with $x = \sigma(1)$ and iterating rightward through σ , we place the values of the permutation into a linear array of “stacks” according to the following rules:

1. Each new value x must be placed at the top of an existing stack, or will form a new stack by itself positioned to the right of all existing stacks.
2. The stack on which each new value x is placed must be the leftmost stack from among those whose current top number is larger than x . If there are no such stacks, then x forms a new stack.

As an example, consider the permutation $\sigma = (4, 1, 2, 7, 6, 5, 8, 9)$. Running the above algorithm on this permutation gives us the following sequence of steps:

$$\begin{array}{cccccccccccccccc}
 4 & \rightarrow & \begin{array}{c} 1 \\ 4 \end{array} & \rightarrow & \begin{array}{c} 1 \quad 2 \\ 4 \end{array} & \rightarrow & \begin{array}{c} 1 \quad 2 \quad 7 \\ 4 \end{array} & \rightarrow & \begin{array}{c} 1 \quad 2 \quad 6 \\ 4 \quad 7 \end{array} & \rightarrow & \begin{array}{c} 1 \quad 2 \quad 5 \\ 4 \quad 6 \\ 7 \end{array} & \rightarrow & \begin{array}{c} 1 \quad 2 \quad 5 \quad 8 \\ 4 \quad 6 \\ 7 \end{array} & \rightarrow & \begin{array}{c} 1 \quad 2 \quad 5 \quad 8 \quad 9 \\ 4 \quad 6 \\ 7 \end{array}
 \end{array}$$

Proposition 2.1.1. *When the patience sorting algorithm is applied to a permutation $\sigma \in S_n$, the final number of stacks is equal to $L(\sigma)$.*

Proof. Let $(\sigma(i_1), \dots, \sigma(i_k))$ be an increasing subsequence of σ of maximal length $k = L(\sigma)$, and let s denote the final number of stacks when the patience sorting algorithm is applied to σ . It is easy to see from the definition of the algorithm that each stack must be decreasing from bottom to top. It follows that $\sigma(i_{j+1})$ must be in a different stack than $\sigma(i_j)$ and consequently that all the $\sigma(i_j)$ must be in different stacks. Therefore, $s \geq L(\sigma)$. Conversely, one can construct an increasing subsequence of σ of length s as follows: let x_s be the top number in the rightmost stack; now let x_{s-1} be the number in the stack immediately to the left that was at the top when x_s was added, and continue in this way down to x_1 . This shows that $s \leq L(\sigma)$. $\square$

Although certainly a useful computational tool, it isn't hard to see that the patience sorting algorithm doesn't offer us any immediate help in understanding the asymptotics of $L(\sigma_n)$, where σ_n denotes a uniformly random permutation of order n . The reason for this is that the patience sorting algorithm records a lot of redundant information, since at any point in the algorithm the only information that is relevant is the numbers at the top of each stack (as it is only these values that play a role in determining where the next value is placed). The true power of the algorithm becomes apparent when it is applied recursively, leading to a procedure known as the **Robinson-Schensted** (RS) algorithm. First, we'll need the concept of a *Young diagram* and *Young tableau*:

Definition 2.1.2. We say that a tuple $\lambda = (\lambda_1, \lambda_2, \dots, \lambda_k)$ of positive integers is a **partition** of n , and write $\lambda \vdash n$, if $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_k$ and $n = \lambda_1 + \dots + \lambda_k$. The integers $\lambda_1, \dots, \lambda_k$ are called the **parts** of the partition. Given a partition $\lambda \vdash n$, we define the **Young diagram** of λ to be a graphical diagram representing the partition λ as a two-dimensional array of boxes, or **cells**, in which the j th row (counting down from the top) has λ_j cells, and the rows are left-justified. A **Young tableau** consists of a Young diagram $\lambda \vdash n$ together with a filling of the cells thereof with the numbers $1, \dots, n$, such that the numbers in each row and each column are arranged in increasing order. We call the diagram of λ the **shape** of the tableau.

We are now ready to state the RS algorithm. Consider a permutation σ of order n . The algorithm proceeds in n steps, defined inductively as follows:

- Begin with a pair of empty tableau (P_0, Q_0) . “Insert” the “biletter” $\left(\begin{smallmatrix} 1 \\ \sigma(1) \end{smallmatrix}\right)$ by adding a box to each of P_0 and Q_0 and filling them with the numbers $\sigma(1)$ and 1, respectively. Call the resulting Young tableau P_1 and Q_1 .

- Having obtained a pair of Young tableau (P_k, Q_k) , insert the biletter $(\sigma(k+1))$ as follows: if the number $\sigma(k+1)$ is larger than or equal to all numbers of the first row of P_k , then we append a box to the end of the first row of P_k and set its content to be $\sigma(k+1)$ and call the resulting tableau P_{k+1} . In this case, we also append a box to the end of the first row of Q_k , set its content to be $k+1$, and call the resulting tableau Q_{k+1} . If, on the other hand, there is a box in the first row of P_k with content strictly larger than $\sigma(k+1)$, then we set the content of the first such box to be $\sigma(k+1)$, and the replaced content (which we shall call b) gets “bumped down” and is row-inserted into the second row of P_k following the same set of rules described thus far, possibly resulting in a cascade of “bump downs”. Eventually, a box is appended either to the end of a row in P_k or below its last row (in which case it creates a new row), and the content of this box will be the last bumped letter. We additionally add a box to Q_k in the same location and set its content to be $k+1$.
- We repeat the above steps until all biletters have been inserted.

Theorem 2.1.1. *The RS algorithm, also called the RS correspondence, is a mapping that takes a permutation $\sigma \in S_n$ and returns a pair (P, Q) of Young tableau of the same shape. This mapping is a bijection between S_n and the set of such pairs. Under this correspondence, we have the identity*

$$L(\sigma) = \text{length of the first row of } P \text{ and } Q.$$

Proof. It is clear that RS defines an injection from S_n to the set of pairs (P, Q) of Young tableau of the same shape. To render it a bijection was the sole purpose of the “recording tableau” Q , which allows us to undo the insertion steps and recover σ , as follows. We know what $\sigma(n)$ must be because it was inserted into P in the final step of the algorithm, and it’s location is given by where n is in the tableau Q . It is furthermore not difficult to see that, by comparing the value of $\sigma(n)$ to the rest of those in P , we can reverse the sequence of bumpings caused by the insertion of $\sigma(n)$ to recover the state of the tableau P and Q prior to $\sigma(n)$ ’s insertion. From here, we can similarly recover the value of $\sigma(n-1)$ as well as the state of tableau P and Q prior to its insertion. By repeating this process, we recover σ entirely. Finally, that $L(\sigma)$ is the length of the first row of P and Q is immediate from Proposition 2.1.1 together with how we defined RS. $\square$

Corollary 2.1.1. *For each $n \in \mathbb{N}$, we have*

$$\sum_{\lambda \vdash n} d_\lambda^2 = n!,$$

where d_λ denotes the number of Young tableau of shape λ . $\square$

The Robinson-Schensted-Knuth (RSK) correspondence is a generalization of the RS correspondence introduced by Knuth, which gives a bijection between matrices with non-negative integer entries and pairs of so-called “semi-standard” Young tableau, which are Young tableau in which the entries of the cells of each row are only required to be *weakly* increasing. To see how it works, note that we can think of an $m \times n$ matrix $W = (w_{i,j})$ as a sequence of m words $w_1, \dots, w_m$, defined as follows:

$$w_i := \underbrace{1 \cdots 1}_{w_{i,1} \text{ times}} \underbrace{2 \cdots 2}_{w_{i,2} \text{ times}} \cdots \underbrace{n \cdots n}_{w_{i,n} \text{ times}}$$

The RSK correspondence consists of applying the RS algorithm to the letters of the words $w_1, \dots, w_m$ (in this order), with the letters of each word w_i being read from left to right.

2.1.3 Gelfand-Tsetlin Parameterization

Besides RSK^{-1} , another useful way of parameterizing Young tableau is the *Gelfand-Tsetlin*, or GT, parameterization. The GT parameterization expresses the entries of a Young tableau as a triangular array of numbers $(z_{i,j})$, $1 \leq j \leq i \leq n$,

$$\begin{array}{ccccccc} & & & & z_{1,1} & & \\ & & & & & z_{2,1} & \\ & & & z_{2,2} & & & \\ & & z_{3,3} & & z_{3,2} & & z_{3,1} \\ & & & \ddots & & \ddots & \\ & & & & \ddots & & \\ z_{n,n} & & z_{n,n-1} & & \cdots & & z_{n,2} & & z_{n,1} \end{array}$$

which “interlace”, by which we mean that

$$z_{i+1,j+1} \leq z_{i,j} \leq z_{i+1,j},$$

by encoding the information of the tableau via

$$z_{i,j} = \sum_{k=1}^i \# \text{ of } k\text{'s in row } j.$$

To see that this definition of $z_{i,j}$ indeed satisfies the interlacing property, notice that the relation $z_{i+1,j+1} \leq z_{i,j}$ is because the entries of the tableau are column-wise strictly increasing, and the other relation is trivial.

2.2 A Solved LPP Model

In this section, we give an example of an LPP model for which an explicit exact solution has been found.

Definition 2.2.1. Given an integral operator $K: L^2(X, \mu) \rightarrow L^2(X, \mu)$, we define its **Fredholm determinant** to be

$$\det(I + K)_{L^2(X, \mu)} := 1 + \sum_{n=1}^{\infty} \frac{1}{n!} \int_{X^n} \det(K(x_i, x_j)) d\mu^n.$$

Note that if K is a trace-class operator, then the above series converges absolutely since it is easily seen to be bounded in modulus by $e^{\text{tr} \sqrt{K^* K}}$.

Remark. To gain some intuition for the choice of notation, note that if K is an $n \times n$ matrix with eigenvalues $\lambda_1, \dots, \lambda_n$, repeated according to multiplicity, then

$$\det(I + K) = \prod_{i=1}^n (1 + \lambda_i) = 1 + \sum_{j=1}^n \sum_{1 \leq i_1 \leq \dots \leq i_j \leq n} \lambda_{i_1} \cdots \lambda_{i_j} = 1 + \sum_{j=1}^n \frac{1}{j!} \sum_{\sigma \in S_j} \det K_{\sigma},$$

where K_{σ} is a $j \times j$ matrix with eigenvalues $\lambda_{i_1} \cdots \lambda_{i_j}$.

Consider an $n \times n$ matrix $W = (w_{(i,j)})$ consisting of random variables $w_{(i,j)}$ with geometric distribution

$$\mathbb{P}(w_{(i,j)} = a) = (1 - p_i q_j)(p_i q_j)^a \mathbf{1}_{a \in \mathbb{Z}_{\geq 0}},$$

where the p_i, q_j are fixed parameters in $(0, 1)$. Let our lattice be $\mathbb{N}^2$ and let $\tau_n := \tau_{(n,n)}$ denote the last passage percolation time of the point (n, n) , so that

$$\tau_n = \max_{\pi \in \Pi_{(n,n)}} \sum_{(i,j) \in \pi} w_{i,j},$$

where, as usual, $\Pi_{(n,n)}$ denotes the set of all up-right paths in $\mathbb{N}^2$ from the origin to the point (n, n) . The content of the following theorem is that the law of τ_n can be computed exactly as a Fredholm determinant.

Theorem 2.2.1. *We have that*

$$\mathbb{P}(\tau_n \leq x) = \det(I + \mathbf{1}_{[x+n, \infty)} K_n^{LPP})_{L^2(\mathbb{N})},$$

where K_n^{LPP} is the operator with kernel

$$K_n^{LPP}(s, t) = \frac{1}{4\pi^2} \int_{\gamma_1} \int_{\gamma_2} \prod_{i=1}^n \left(\frac{1 - p_i \zeta}{t - q_i} \right) \prod_{j=1}^n \left(\frac{1 - \eta q_j}{\eta - p_j} \right) \cdot \frac{\eta^t \zeta^s}{1 - \zeta \eta} d\eta d\zeta.$$

Above, γ_1 denotes the counter-clockwise oriented circle of radius $r < 1$ about the origin and γ_2 the counter-clockwise oriented unit circle.

While seemingly not so nice upon first glance, asymptotic analysis of the above formula is actually not so difficult, and can be done via the so-called *method of steepest descent*. The idea is as follows. Suppose one is interested in the behavior of

$$\int_I f(x)e^{ng(x)}dx$$

as $n \rightarrow \infty$, where f, g are real-valued analytic functions and I is some open interval in $\mathbb{R}$. If g has a unique maximum at $x_0 \in I$, then the main contribution to the above comes from a neighbourhood of x_0 . In particular, if $g''(x_0) < 0$, then by Taylor expanding g to the second order about x_0 we get

$$g(x) = g(x_0) + \frac{1}{2}g''(x_0)(x - x_0)^2 + o((x - x_0)^2)$$

and it follows that, for large n ,

$$\int_I f(x)e^{ng(x)}dx \approx f(x_0)e^{ng(x_0)} \int_{x_0-\varepsilon}^{x_0+\varepsilon} e^{n[\frac{1}{2}g''(x_0)+o((x-x_0)^2)]}dx \approx f(x_0)e^{ng(x_0)} \left(\frac{2\pi}{-ng''(x_0)} \right)^{1/2}.$$

Clearly, the latter expression is much easier to work with than what we started with. In our case, we wish to analyze something like

$$\int_{\gamma} f(z)e^{ng(z)}dz$$

as $n \rightarrow \infty$, where f and g are analytic functions of a complex variable and γ is a complex contour. The idea is to reduce to the case above by transforming the contour γ (via Cauchy's theorem) so that it passes through the critical points of g in such a way that the contribution to the above away from the critical points is negligible.

Chapter 3

Longest Increasing Subsequences in Random Permutations

The Hammersley problem of the previous chapter motivates the study of longest increasing subsequences of random permutations. For context, if $\sigma \in S_n$ is a permutation, we define a **subsequence** of σ to be any sequence $(\sigma(i_1), \dots, \sigma(i_k))$, where $1 \leq i_1 < \dots < i_k \leq n$. We say that such a subsequence is **increasing** if $\sigma(i_1) < \dots < \sigma(i_k)$, **decreasing** if $\sigma(i_1) > \dots > \sigma(i_k)$, and **monotone** if it is either increasing or decreasing. Given a permutation σ chosen uniformly at random among all permutations of order n , the goal of this chapter is to conduct a detailed study of the asymptotics of *the length of the longest increasing subsequence of σ* .

3.1 The Ulam-Hammersley Problem

Given a permutation $\sigma \in S_n$, we let $L(\sigma)$ denote the length of the longest increasing subsequence of σ ; that is,

$$L(\sigma) := \max\{1 \leq k \leq n : \sigma \text{ has an increasing subsequence of length } k\}.$$

By considering the average value of $L(\sigma)$ as σ ranges over all $n!$ permutations of order n , we get a natural sequence

$$(\ell_1, \ell_2, \ell_3, \dots, \ell_n, \dots), \quad \text{where} \quad \ell_n = \frac{1}{n!} \sum_{\sigma \in S_n} L(\sigma).$$

The *Ulam-Hammersley problem* is to determine the asymptotic behavior of ℓ_n as $n \rightarrow \infty$. Alongside developing an understanding of how $L(\sigma_n)$ behaves as $n \rightarrow \infty$ (where σ_n denotes a uniformly random permutation of order n), this is the goal of this chapter.

The first major breakthrough in understanding the asymptotics of ℓ_n came from a 1935 result of Paul Erdos and George Szekeres, which roughly asserts that if a permutation has no long increasing subsequence, then it must have a compensatory long *decreasing* subsequence. More precisely,

Theorem 3.1.1 (Erdos-Szekeres Theorem). *If $\sigma \in S_n$ and $n > rs$ for some choice of positive integers $r, s \in \mathbb{N}$, then $L(\sigma) > r$ or $D(\sigma) > s$.*

Proof. For each $1 \leq k \leq n$, let $L_k(\sigma)$ denote the maximal length of an increasing subsequence of σ which ends with $\sigma(k)$, and similarly let $D_k(\sigma)$ denote the maximal length of a decreasing subsequence of σ which ends with $\sigma(k)$.

Consider, now, the n pairs $(D_k(\sigma), L_k(\sigma))$, $1 \leq k \leq n$. Note that these pairs must be distinct from one another. For if $1 \leq j < k \leq n$, then $\sigma(j) \neq \sigma(k)$; if $\sigma(j) < \sigma(k)$, then $L_j(\sigma) < L_k(\sigma)$ since we can construct an increasing subsequence of σ ending in $\sigma(k)$ of length larger than $L_j(\sigma)$ by first taking an increasing subsequence of σ ending in $\sigma(j)$ of length $L_j(\sigma)$ and appending $\sigma(k)$ to it. Similarly, we obtain that $D_j(\sigma) < D_k(\sigma)$ if $\sigma(j) > \sigma(k)$.

It follows that for some $1 \leq k \leq n$ we have $L(\sigma) \geq L_k(\sigma) > r$ or $D(\sigma) \geq D_k(\sigma) > s$, for otherwise the n distinct pairs $(D_k(\sigma), L_k(\sigma))$ would all lie in the set $\{1, \dots, r\} \times \{1, \dots, s\}$, contradicting the hypothesis that $n > rs$. $\square$

A useful re-formulation of this theorem (which perhaps renders its underlying meaning more clear) is the following:

Corollary 3.1.1. *If $\sigma \in S_n$, then $L(\sigma)D(\sigma) \geq n$.*

Proof. Let $r = L(\sigma)$ and $s = D(\sigma)$. If $rs < n$, then by the above theorem we have that $r = L(\sigma) > r$ or $s = D(\sigma) > s$; however, both of these conclusions are clearly absurd. Therefore, $rs \geq n$. $\square$

The significance of this result to the problem we're interested in is that it gives rise to a lower bound for ℓ_n . In particular,

Proposition 3.1.1. *For all $n \geq 1$ we have that $\ell_n \geq \sqrt{n}$.*

Proof. Recall that ℓ_n is the average value of $L(\sigma)$ over all $\sigma \in S_n$. Because each permutation σ has a corresponding “reverse” permutation τ (defined by $\sigma(i) = j$ if and only if $\tau(n - i + 1) = j$) in which all increasing subsequences become decreasing and vice-versa, ℓ_n is also the average value of $D(\sigma)$ over all $\sigma \in S_n$. This means that we can write

$$\ell_n = \frac{1}{n!} \sum_{\sigma \in S_n} \frac{L(\sigma) + D(\sigma)}{2}.$$

Re-phrasing in terms of the language of probability, note that if σ_n denotes a uniformly random permutation of order n , then $\ell_n = \mathbb{E}L(\sigma_n) = \mathbb{E}D(\sigma_n)$ so by linearity of expectations of random variables we have

$$\ell_n = \mathbb{E} \left(\frac{L(\sigma_n) + D(\sigma_n)}{2} \right).$$

It remains to note that since $(L(\sigma_n) + D(\sigma_n))/2 \geq \sqrt{L(\sigma_n)D(\sigma_n)}$ by the AM-GM inequality, monotonicity of expectations and [Corollary 3.1.1](#) imply that

$$\ell_n \geq \mathbb{E}(\sqrt{L(\sigma_n)D(\sigma_n)}) \geq \sqrt{n},$$

as claimed. $\square$

The content of the following proposition is a corresponding upper bound for ℓ_n (at least for sufficiently large n):

Proposition 3.1.2.

$$\limsup_{n \rightarrow \infty} \frac{\ell_n}{\sqrt{n}} \leq e.$$

Proof. Let σ_n denote a uniformly random permutation of order n , and for each $1 \leq k \leq n$ define $X_{n,k}$ to be the number of increasing subsequences of σ_n of length k . Then

$$\mathbb{E}X_{n,k} = \frac{1}{k!} \binom{n}{k}$$

since there are $\binom{n}{k}$ subsequences of length k , and the probability that any given one is increasing is $1/k!$. It follows that

$$\mathbb{P}(L(\sigma_n) \geq k) = \mathbb{P}(X_{n,k} \geq 1) \leq \mathbb{E}X_{n,k} = \frac{n(n-1) \cdots (n-k+1)}{(k!)^2} \leq \frac{n^k}{(k/e)^{2k}}.$$

Above, the first inequality comes from the fact that

$$\mathbb{P}(X_{n,k} \geq 1) = \int \mathbf{1}_{\{x: X_{n,k}(x) \geq 1\}} d\mathbb{P}(x) \leq \int X_{n,k}(x) d\mathbb{P}(x) = \mathbb{E}X_{n,k}$$

and the final equality follows from Stirling's approximation: $n! = (1 + o(1))\sqrt{2\pi n}(n/e)^n$. If $\delta > 0$ is any positive number, taking $k = (1 + \delta)e\sqrt{n}$ then shows that

$$\mathbb{P}(L(\sigma_n) \geq k) \leq \frac{n^k}{(k/e)^{2k}} \leq \left(\frac{1}{1 + \delta} \right)^{2k} = \left(\frac{1}{1 + \delta} \right)^{2(1+\delta)e\sqrt{n}}$$

Therefore,

$$\begin{aligned}\ell_n = \mathbb{E}L(\sigma_n) &= \int L(\sigma_n) d\mathbb{P} = \int_{\{x: L(\sigma_n) < k\}} L(\sigma_n) d\mathbb{P} + \int_{\{x: L(\sigma_n) \geq k\}} L(\sigma_n) d\mathbb{P} \\ &\leq k\mathbb{P}(L(\sigma_n) < k) + n\mathbb{P}(L(\sigma_n) \geq k) \\ &\leq (1 + \delta)e\sqrt{n} + O(e^{-c\sqrt{n}}),\end{aligned}$$

where c is some positive constant that depends on δ . Dividing by $\sqrt{n}$, we get

$$\frac{\ell_n}{\sqrt{n}} \leq (1 + \delta)e + O(e^{-c\sqrt{n}}/\sqrt{n}),$$

whence $\limsup_{n \rightarrow \infty} \ell_n/\sqrt{n} \leq (1 + \delta)e$. As $\delta > 0$ was arbitrary, the result follows. $\square$

Corollary 3.1.2. *For any $\alpha > e$, we have that*

$$\mathbb{P}(L(\sigma_n) > \alpha\sqrt{n}) \leq Ce^{-c\sqrt{n}}$$

for some constants $C, c > 0$ which depend on α but not on n . In particular, $e\sqrt{n}$ is not only an asymptotic bound for ℓ_n , the average value of $L(\sigma)$ over all permutations σ of order n , but also an asymptotic bound for the **typical** value of $L(\sigma_n)$.

Proof. Recall from the proof of the above proposition that we have the estimate

$$\mathbb{P}(L(\sigma_n) \geq (1 + \delta)e\sqrt{n}) \leq \left(\frac{1}{1 + \delta}\right)^{2(1 + \delta)e\sqrt{n}} = e^{-2(1 + \delta)\log(1 + \delta)e\sqrt{n}},$$

valid for any $\delta > 0$. Clearly, this is equivalent to

$$\mathbb{P}(L(\sigma_n) \geq \alpha\sqrt{n}) \leq e^{-2\alpha\log(\alpha/e)\sqrt{n}}$$

for any $\alpha > e$. The result therefore follows by taking $c = 2\alpha\log(\alpha/e)$, $C = 1$, and noting that $\mathbb{P}(L(\sigma_n) > \alpha\sqrt{n}) \leq \mathbb{P}(L(\sigma_n) \geq \alpha\sqrt{n})$. The final remark is because it follows that $\mathbb{P}(L(\sigma_n) > (1 + \delta)e\sqrt{n}) \rightarrow 0$ as $n \rightarrow \infty$ for any $\delta > 0$, and therefore that $\mathbb{P}(L(\sigma_n) \leq e\sqrt{n}) \rightarrow 1$ as $n \rightarrow \infty$. $\square$

The next major breakthrough we discuss was due to Hammersley, and roughly states that the behavior of $L(\sigma_n)$ for a typical permutation σ_n is asymptotically equivalent to the behavior of ℓ_n , its average value. This is non-obvious because there is no reason to suspect that pathological cases such as half of the permutations having value $L(\sigma)$ close to 0 and the other half having value $L(\sigma)$ close to $c\sqrt{n}$ for some suitable constant c are impossible. The precise result is the following:

Theorem 3.1.2 (Hammersley's Theorem). *The limit $\Lambda = \lim_{n \rightarrow \infty} \ell_n/\sqrt{n}$ exists. Moreover, $L(\sigma_n)/\sqrt{n} \rightarrow \Lambda$ in probability as $n \rightarrow \infty$.*

It is an immediate corollary of [Proposition 3.1.1](#) and [Proposition 3.1.2](#) that $1 \leq \Lambda \leq e \approx 2.718$. It can in fact be shown that $\Lambda = 2$; meaning that, for large n , $\ell_n \approx 2\sqrt{n}$ and $L(\sigma_n) \approx 2\sqrt{n}$ with high probability.

3.2 The Hook-Length Formula

The RS correspondence provides us with a useful way of reformulating our questions about the asymptotic behavior of $L(\sigma_n)$. Namely, when applying the RS algorithm to σ_n , one obtains a random Young diagram $\lambda^{(n)}$ (diagram, as opposed to tableau, since the content of the entries isn't actually necessary for computing $L(\sigma_n)$, so we may as well forget about them), the length of the first row of which equals $L(\sigma_n)$ (by [Theorem 2.1.1](#)).

To determine how $\lambda^{(n)}$ behaves problematically speaking, we note that because σ_n is a uniformly random permutation of order n , the probability that $\lambda^{(n)} = \lambda$ is $1/n!$ times the number of permutations $\sigma \in S_n$ whose RS shape is λ . By the algorithm, the set of such permutations is in bijection with the set of all pairs (P, Q) of Young tableau of shape λ . Letting d_λ denote the number of Young tableau of shape λ , then, we have

$$\mathbb{P}(\lambda^{(n)} = \lambda) = \frac{d_\lambda^2}{n!}.$$

That the quantities on the right-hand side sum to 1 is an immediate consequence of [Corollary 2.1.1](#). The probability measure on the set of integer partitions of n given by the above (i.e. which assigns probability $d_\lambda^2/n!$ to any partition λ) is known as the *Plancherel measure* of order n .

A good way of analyzing the Plancherel measure comes from the remarkable *Hook-Length formula*, which is the principle result of this section. If (i, j) denotes a cell in the Young diagram of a partition $\lambda \vdash n$ (where i is the row index and j the column index), we define the **arm** of (i, j) to be the collection of cells in λ of the form (i, k) , where

$$j \leq k \leq \lambda_i := \text{length of } i\text{th row of } \lambda;$$

that is, the cell (i, j) together with all cells in the same row as (i, j) which lie to its right. Analogously, we define the **leg** of (i, j) to be the collection of all cells in λ of the form (k, j) for

$$i \leq k \leq \lambda'_j := \text{length of } j\text{th column of } \lambda;$$

that is, the cell (i, j) together with all cells in the same column in (i, j) which lie below it. We define the **hook** of (i, j) to be the union of its arm and its leg, and denote the set of such cells by $H_\lambda(i, j)$. Finally, we define the **hook length** of (i, j) to be the number $h_\lambda(i, j)$ of cells in its hook. It is easy to see from how we defined the hook of a cell that

$$h_\lambda(i, j) = \lambda_i - j + \lambda'_j - i + 1.$$

Theorem 3.2.1 (The Hook Length Formula). *If $\lambda \vdash n$, then we have*

$$d_\lambda = \frac{n!}{\prod_{(i,j)} h_\lambda(i, j)}. \quad (3.1)$$

Proof. Let us denote the right-hand side of [Eq. \(3.1\)](#) by e_λ , so that our goal is to show that $d_\lambda = e_\lambda$ for all integer partitions λ . We first notice that d_λ satisfies a recurrence relation:

$$d_\lambda = \sum_{\mu \nearrow \lambda} d_\mu,$$

where we use the notation $\mu \nearrow \lambda$ to indicate that the Young diagram of λ can be obtained from the Young diagram of μ by the addition of a single cell. This is because for any given Young tableau λ , deleting the maximal entry yields a tableau of shape μ for some $\mu \nearrow \lambda$. Together with the “initial condition” $d_\emptyset = 1$ (for the “empty” diagram of size 0), this recurrence determines all values of d_λ uniquely. Since it is trivial that $e_\emptyset = 1$, it shall suffice to show that e_λ satisfies the same recurrence; that is,

$$e_\lambda = \sum_{\mu \nearrow \lambda} e_\mu \iff 1 = \sum_{\mu \nearrow \lambda} \frac{e_\mu}{e_\lambda}.$$

To do this, we shall construct a probability measure on the collection of diagrams μ satisfying $\mu \nearrow \lambda$ which assigns a probability of e_μ/e_λ to each such μ :

We first note that the set of μ such that $\mu \nearrow \lambda$ are clearly in bijection with the cells (i, j) that can be removed from λ to obtain a new valid Young diagram μ (here, we mean literally removed, and not removed in the sense of a deletion step). We refer to such cells as the “corners” of λ ; they are the cells which are simultaneously the last in their row and in their column, or equivalently, the cells that are the only elements of their hook.

We now fix a diagram λ and consider the following probabilistic process, which we’ll call a “hook walk,” that leads to a random choice of corner. We begin by selecting an initial cell for the walk to begin on, uniformly at random out of the $|\lambda|$ possibilities. Next, we perform a random walk, successively replacing the current cell with a new one until a corner is reached, according to the rule that the successor of the current cell (i, j) must be chosen uniformly at random from the hook $H_\lambda(i, j) \setminus \{(i, j)\}$. Because corners are the only cells in their hooks, the walk terminates once we’ve reached a corner. If we denote this terminating corner by $\mathbf{c}$, then we have reduced our original problem to showing that for any corner cell (a, b) of λ we have that

$$\mathbb{P}(\mathbf{c} = (a, b)) = \frac{e_\mu}{e_\lambda}, \quad \text{where } \mu = \lambda \setminus \{(a, b)\}. \quad (3.2)$$

To do *this*, we first let $n = |\lambda|$ and note that the right-hand side can be written as

$$\frac{e_\mu}{e_\lambda} = \frac{(n-1)! / \prod_{(i,j)} h_\mu(i, j)}{n! / \prod_{(i,j)} h_\lambda(i, j)} = \frac{1}{n} \prod_{(i,j)} \frac{h_\lambda(i, j)}{h_\mu(i, j)}.$$

Since $\mu \nearrow \lambda$, we next note that in the above product all terms cancel out except those in the same row or column as the corner cell (a, b) . Thus, the above simplifies to

$$\begin{aligned} \frac{e_\mu}{e_\lambda} &= \frac{1}{n} \prod_{i=1}^{a-1} \frac{h_\lambda(i, b)}{h_\lambda(i, b) - 1} \prod_{j=1}^{b-1} \frac{h_\lambda(a, j)}{h_\lambda(a, j) - 1} \\ &= \frac{1}{n} \prod_{i=1}^{a-1} \left(1 + \frac{1}{h_\lambda(i, b) - 1}\right) \prod_{j=1}^{b-1} \left(1 + \frac{1}{h_\lambda(a, j) - 1}\right) \\ &= \frac{1}{n} \sum_{A \subseteq \{1, \dots, a-1\}} \sum_{B \subseteq \{1, \dots, b-1\}} Q(A, B), \end{aligned}$$

where $Q(A, B) = \prod_{i \in A} \frac{1}{h_\lambda(i, b) - 1} \prod_{j \in B} \frac{1}{h_\lambda(a, j) - 1}$.

We now give a probabilistic interpretation of $Q(A, B)$ that shall link it to the left-hand side of Eq. (3.2). For a possible trajectory $(x_1, y_1) \rightarrow \dots \rightarrow (x_k, y_k)$ of cells visited by the hook walk, let us define its horizontal and vertical projections to be the sets $\{y_1, \dots, y_k\}$ and $\{x_1, \dots, x_k\}$, respectively. In other words, the horizontal projection of trajectory is simply the set of columns it visits, and the vertical projection the set of rows it visits.

Now, for sets $A \subseteq \{1, \dots, a-1\}$ and $B \subseteq \{1, \dots, b-1\}$, let us define a quantity $P(A, B)$ to be the conditional probability for a hook walk, conditioned to start from the initial cell $(\min A \cup \{a\}, \min B \cup \{b\})$, to reach the corner (a, b) while passing through a trajectory $(x_1, y_1) \rightarrow \dots \rightarrow (x_k, y_k) = (a, b)$ whose vertical projection is $A \cup \{a\}$ and whose horizontal projection is $B \cup \{b\}$. Put differently, *given* a hook walk starting from initial cell $(\min A \cup \{a\}, \min B \cup \{b\})$, $P(A, B)$ is the probability that this walk reaches the corner (a, b) while visiting all the rows prescribed by A and all the columns prescribed by B .

We claim that $P(A, B) = Q(A, B)$. Note that this will complete the proof, for we can then expand the left-hand side of Eq. (3.2) according to the initial cell (x_1, y_1) to get

$$\begin{aligned} \mathbb{P}(\mathbf{c} = (a, b)) &= \frac{1}{n} \sum_{x_1=1}^a \sum_{y_1=1}^b \sum_{\substack{H \subseteq \{x_1, \dots, a\} \\ \min H = x_1, \max H = a}} \sum_{\substack{V \subseteq \{y_1, \dots, b\} \\ \min V = y_1, \max V = b}} P(H \setminus \{a\}, V \setminus \{b\}) \\ &= \frac{1}{n} \sum_{x_1=1}^a \sum_{y_1=1}^b \sum_{\substack{H \subseteq \{x_1, \dots, a\} \\ \min H = x_1, \max H = a}} \sum_{\substack{V \subseteq \{y_1, \dots, b\} \\ \min V = y_1, \max V = b}} Q(H \setminus \{a\}, V \setminus \{b\}), \end{aligned}$$

and a few moments of reflection show that this is equal to $\frac{1}{n} \sum_{A \subseteq \{1, \dots, a-1\}} \sum_{B \subseteq \{1, \dots, b-1\}} Q(A, B)$, which we showed previously to be e_μ/e_λ . To show that $P(A, B) = Q(A, B)$, we proceed by induction on the length k of the hook walk. The base case $k = 1$ corresponds to the case $A = B = \emptyset$ and is trivial. Suppose now that $k > 1$, so that at least one of A or B is non-empty. In fact, let us suppose that both A and B are non-empty (the case where one of them is empty is similar). Letting $i_1 = \min A$ and $j_1 = \min B$, we have

$$P(A, B) = \frac{1}{h_\lambda(i_1, j_1) - 1} [P(A \setminus \{i_1\}, B) + P(A, B \setminus \{j_1\})]$$

(this is because when we move from our initial cell (i_1, j_1) , we must move to a new cell in the hook of (i_1, j_1) uniformly at random, each one of which is either in a different row than i_1 or a different column than j_1). By our inductive hypothesis, this is equal to

$$\begin{aligned} &\frac{1}{h_\lambda(i_1, j_1) - 1} [Q(A \setminus \{i_1\}, B) + Q(A, B \setminus \{j_1\})] \\ &= \frac{1}{h_\lambda(i_1, j_1) - 1} \left(\prod_{i \in A \setminus \{i_1\}} \frac{1}{h_\lambda(i, b) - 1} \prod_{j \in B} \frac{1}{h_\lambda(a, j) - 1} + \prod_{i \in A} \frac{1}{h_\lambda(i, b) - 1} \prod_{j \in B \setminus \{j_1\}} \frac{1}{h_\lambda(a, j) - 1} \right) \\ &= \frac{1}{h_\lambda(i_1, j_1) - 1} \prod_{i \in A \setminus \{i_1\}} \frac{1}{h_\lambda(i, b) - 1} \prod_{j \in B \setminus \{j_1\}} \frac{1}{h_\lambda(a, j) - 1} \left(\frac{1}{h_\lambda(a, j_1) - 1} + \frac{1}{h_\lambda(i_1, b) - 1} \right) \\ &= \frac{[h_\lambda(i_1, b) - 1] + [h_\lambda(a, j_1) - 1]}{h_\lambda(i_1, j_1) - 1} \prod_{i \in A} \frac{1}{h_\lambda(i, b) - 1} \prod_{j \in B} \frac{1}{h_\lambda(a, j) - 1} \\ &= Q(A, B), \end{aligned}$$

where the last equality follows since

$$\begin{aligned}
[h_\lambda(i_1, b) - 1] + [h_\lambda(a, j_1) - 1] &= (\lambda_{i_1} - b + \lambda'_b - i_1) + (\lambda_a - j_1 + \lambda'_{j_1} - a) \\
&= (\lambda_{i_1} - j_1 + \lambda'_{j_1} - i_1) + (\lambda_a - b + \lambda'_b - a) \\
&= (h_\lambda(i_1, j_1) - 1) + (h_\lambda(a, b) - 1) \\
&= h_\lambda(i_1, j_1) - 1.
\end{aligned}$$

This completes the proof of the hook length formula. $\square$

3.3 Asymptotic Analysis of the Hook-Length Formula

When one generates a Plancherel-random partition $\lambda^{(n)}$ and applies the RS algorithm to it, it is apparent that the corresponding Young diagram converges to a continuous shape. One can run the simulation many times and still get a shape that looks approximately the same. It turns out that this is no coincidence, and rather an instance of a *limit shape* phenomenon; that is, the phenomenon of some geometric object defined on a class of random discrete objects converging to a limiting continuous shape as the size parameter tends to infinity. The goal of the next two sections will be to rigorously prove the existence of this limit shape (in an appropriate limiting sense), and to find an explicit formula for it.

Since we will be talking about convergence of a sequence of Young diagrams to a limiting shape, we shall need to define the space of shapes on which our analysis will take place, as well as our notion of convergence on this space. Since integer partitions are described by a non-increasing sequence of non-negative integers that sum to a given integer n , we opt to describe the analogous limiting objects as non-negative, non-increasing functions on the positive real-line that integrate to a given real number, say (for convenience) 1. Let us define $\mathcal{F}$ to be the set of such functions; that is,

$$\mathcal{F} := \{f: [0, \infty) \rightarrow [0, \infty) : f \text{ non-increasing, } \int_0^\infty f(x)dx = 1, f \text{ has compact support}\}.$$

The requirement that our shapes be compactly supported is also made for convenience, and not an obvious one to include as there is no reason a priori to suspect that the limit shape should be compactly supported. As we shall see, however, it does simplify the analysis considerably without affecting the generality of our results. We shall occasionally refer to elements of $\mathcal{F}$ as **continual Young diagrams**.

Given that $\mathcal{F}$ is to be the space of shapes over which analysis is to occur, we must be able to embed Young diagrams into $\mathcal{F}$ somehow. The natural way to do this is by associating to a Young diagram λ of order n the function $\phi_\lambda \in \mathcal{F}$ defined by

$$\phi_\lambda(x) := n^{-1/2} \lambda'_{\lfloor \sqrt{n}x \rfloor + 1}.$$

In other words, we represent the Young diagram λ by the continuous function which outlines it upon placing it in the northeast quadrant so that its first row is aligned with the y -axis, and then scaling both the horizontal and vertical axes by a factor of $\sqrt{n}$ so that the area of the Young diagram shrinks from n to 1. For now, we defer the decision as to a topology on $\mathcal{F}$; the right choice shall become apparent later in our analysis.

The first thing we endeavour to do is to find an analytical expression that appears in connection with functions f in our space of shapes $\mathcal{F}$, and which can be used to get a very good approximation of the hook length product appearing in $d_\lambda^2/n!$ (as per the hook length formula) for large Young diagrams.

Given $f \in \mathcal{F}$ and a point $(x, y) \in [0, \infty) \times [0, \infty)$ under the graph of f , we (analogously to Young diagrams) define the **arm** of (x, y) to be the set

$$\{(x', y) : x \leq x' \leq f^{-1}(y)\}$$

(here, $f^{-1}(y) := \inf\{x \geq 0 : f(x) \leq y\}$) and the **leg** of (x, y) to be the set

$$\{(x, y') : y \leq y' \leq f(x)\}.$$

We then define the **hook** of (x, y) , denoted $H_f(x, y)$ to be the union of the arm and the leg of (x, y) , and the **hook length** of (x, y) , denoted $h_f(x, y)$, to be the length of $H_f(x, y)$; that is,

$$h_f(x, y) = f(x) - y + f^{-1}(y) - x,$$

where $f^{-1}(y)$ is again defined as above. We furthermore define the **hook integral** of f via the expression

$$I_{\text{hook}}(f) = \int_0^\infty \int_0^{f(x)} \log h_f(x, y) dy dx.$$

The hook integral is related to the hook length formula and the Plancherel measure via the following result:

Theorem 3.3.1 (Asymptotic Hook Length Formula). *As $n \rightarrow \infty$, we have, uniformly over all partitions λ of n , that*

$$\frac{d_\lambda^2}{n!} = \exp \left[-n \left(1 + 2I_{\text{hook}}(\phi_\lambda) + O\left(\frac{\log n}{\sqrt{n}}\right) \right) \right].$$

3.4 The Variational Problem for Hook Integrals

By Theorem 3.3.1, minimizing $I_{\text{hook}}(\phi_\lambda)$ has the effect of maximizing the probability that ϕ_λ arises as the shape of a Plancherel-random partition of high order. Since such a shape is obviously a good candidate for the limiting shape, we naturally arrive at the *variational problem for hook integrals*: find a function $f \in \mathcal{F}$ for which $I_{\text{hook}}(f)$ is minimized, and compute the value of $I_{\text{hook}}(f)$.

As it currently stands, the hook integral formula is difficult to analyze as it involves the expression $h_f(x, y)$, which we don't have a nice formula for. To make it simpler, let us make the change of coordinates

$$u = \frac{x - y}{\sqrt{2}}, \quad v = \frac{x + y}{\sqrt{2}}$$

which rotates our perspective by 45 degrees. Let $\mathcal{G}$ denote the set of continual Young diagrams $\mathcal{F}$ drawn in this new coordinate system. It is straightforward to verify that $\mathcal{G}$ is the exactly the space of functions $g: \mathbb{R} \rightarrow [0, \infty)$ satisfying the following properties:

1. g is 1-Lipschitz; i.e., $|g(s) - g(t)| \leq |s - t|$ for all $s, t \in \mathbb{R}$,
2. $g(u) \geq |u|$ for all $u \in \mathbb{R}$,
3. $\int_{-\infty}^\infty g(u) - |u| du = 1$,
4. $g(u) - |u|$ is compactly supported.

Note that the domain of integration for the hook integral is the region under the graph of f . Equivalently, in the language of the uv -coordinate system and our transformed space of functions $\mathcal{G}$, this is the region between g and the graph of $v = |u|$. We now define a second transformation which parameterizes this region in terms of another set of coordinates called “hook coordinates”, which are defined as follows: if (u, v) is a point such that $|u| \leq v \leq g(u)$, consider the hook of the point (u, v) ; denoting the points of intersection between this hook and the graph of g by $(s, g(s))$ and $(t, g(t))$, the hook coordinates of (u, v) are given by (t, s) . An equivalent, more formal definition of t and s is as solutions to the following system of equations:

$$g(s) - s = v - u, \quad g(t) + t = v + u. \tag{3.3}$$

Finally, if we denote by Δ the region in the st -plane to which the region between the graphs of g and $v = |u|$ is mapped via the hook coordinate transformation, then we can re-express the hook integral as an integral over Δ as follows. In the st -coordinate system, the pesky $\log h_f(x, y)$ term becomes $\log \sqrt{2}(s - t)$, so we have

$$I_{\text{hook}}(f) = \int_{\Delta} \log \sqrt{2}(s - t) |\det D((t, s) \mapsto (u, v))| ds dt,$$

by the Change of Variables formula. Now, using the relations in Eq. (3.3), one can solve for $\partial s / \partial u$, $\partial s / \partial v$, $\partial t / \partial u$, and $\partial t / \partial v$ to get that

$$\frac{\partial t}{\partial u} = \frac{\partial t}{\partial v} = \frac{1}{1 + g'(t)}, \quad \text{and} \quad \frac{\partial s}{\partial u} = -\frac{\partial s}{\partial v} = \frac{1}{1 - g'(s)}.$$

It follows that

$$\det D((u, v) \mapsto (t, s)) = \det \begin{pmatrix} \partial t / \partial u & \partial t / \partial v \\ \partial s / \partial u & \partial s / \partial v \end{pmatrix} = \frac{2}{(g'(s) - 1)(g'(t) + 1)}$$

and consequently that

$$I_{\text{hook}}(f) = \frac{1}{2} \int_{\Delta} \log \sqrt{2}(s-t)(1+g'(t))(1-g'(s))dsdt = \frac{1}{2} \int_{t < s} \log \sqrt{2}(s-t)(1+g'(t))(1-g'(s))dsdt,$$

where the last equality is because, outside Δ , we either have that $1+g'(t) = 0$ or $1-g'(s) = 0$ (since eventually we have that $g(t) = -t$ and $g(t) = t$ by compact support). To simplify the above, note that by writing $\log \sqrt{2}(s-t)$ as $\log \sqrt{2} + \log(s-t)$ and expanding the integral we get

$$\begin{aligned} I_{\text{hook}}(f) &= \frac{1}{2} \int_{\Delta} \log \sqrt{2}(1+g'(t))(1-g'(s))dsdt + \frac{1}{2} \int_{t < s} \log(s-t)(1+g'(t))(1-g'(s))dsdt \\ &= \frac{1}{2} \log 2 + \frac{1}{2} \int_{t < s} \log(s-t)(1+g'(t))(1-g'(s))dsdt. \end{aligned}$$

Next, let us define $h(u) := g(u) - |u|$, so that $h'(u) = g'(u) - \text{sgn}(u)$, and observe that it is convenient to write the integral $\int_{t < s} \log(s-t)(1+g'(t))(1-g'(s))dsdt$ as

$$- \int_{t < s} \log(s-t)h'(t)h'(s)dtds + \left(\frac{1}{2} \int_{t < s} \log(s-t)(1+g'(t))(1-g'(s))dsdt + \int_{t < s} \log(s-t)h'(t)h'(s)dtds \right)$$

and the second term equals

$$\int_{t < s} \log(s-t)(g'(t) - \text{sgn}(t))(\text{sgn}(s) - g'(s))dtds.$$

This expression is useful because if we then split the domain of integration into parts according to the sign of t and s , then the part corresponding to $t < 0 < s$ disappears due to cancellation and we are left with

$$\begin{aligned} \int_{t < s} \log(s-t)(1+g'(t))(1-g'(s))dsdt &= 2 \int_{t < s < 0} \log(s-t)h'(t) - 2 \int_{0 < t < s} \log(s-t)h'(s)dtds \\ &= 2 \int_{-\infty}^0 h'(t) \left(\int_t^0 \log(s-t)ds \right) dt - 2 \int_0^{\infty} h'(s) \left(\int_0^s \log(s-t)dt \right) ds \\ &= -2 \int_{-\infty}^{\infty} h'(u)(u \log |u| - u)du. \end{aligned}$$

If we then define operators Q and L via

$$Q(h) := -\frac{1}{2} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \log |s-t| h'(t)h'(s)dtds, \quad L(h) := -2 \int_{-\infty}^{\infty} h'(u)(u \log |u| - u)du,$$

then it follows that

$$2I_{\text{hook}}(f) = \log 2 + Q(h) + L(h).$$

If we define the right-hand side to be $J(h)$, then we see that the problem of minimizing I_{hook} is reduced to minimizing $J(h)$ over all suitable functions h , which turns out to be the set $\mathcal{H}$ of functions $h: \mathbb{R} \rightarrow [0, \infty)$ with the following properties:

1. h is absolutely continuous,
2. $\int_{-\infty}^{\infty} h(x)dx = 1$,
3. $\text{sgn}(x) \cdot h'(x) \in [-2, 0]$ wherever defined,
4. h is compactly supported.

Definition 3.4.1. Given a suitable (e.g. L^2) function $f: \mathbb{R} \rightarrow \mathbb{R}$, we define its **Hilbert transform** to be the function $\tilde{f}$ defined by

$$\tilde{f}(x) = \frac{1}{\pi} \int_{-\infty}^{\infty} \frac{f(y)}{x-y} dy,$$

where the integral is required to converge almost everywhere in the sense of the Cauchy Principle Value. Moreover, we define its **Fourier transform** to be the function $\hat{f}$ defined by

$$\hat{f}(x) = \int_{-\infty}^{\infty} f(u)e^{-ixu} du.$$

It is well-known fact that when $\tilde{f}$ and $\widehat{f}$ exist, we have the inverse relations

$$f(x) = -\frac{1}{\pi} \int_{-\infty}^{\infty} \frac{\tilde{f}(y)}{x-y} dy, \quad f(x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \widehat{f}(u) e^{ixu} du.$$

Lemma 3.4.1. *For any compactly supported Lipschitz function $h: \mathbb{R} \rightarrow \mathbb{R}$, we have that $Q(h) \geq 0$ with equality if and only if $h \equiv 0$.*

Proof. Let h be as above. By the Lipschitz property, it is straightforward to verify that $\tilde{h}$ exists. Moreover, by the compact support property it is easy to see that $\widehat{h}$ exists. Integrating by parts, we note that $\tilde{h}$ can equivalently be expressed as

$$\tilde{h}(x) = \frac{1}{\pi} \int_{-\infty}^{\infty} h'(u) \log|x-u| du.$$

In turn, it follows that

$$Q(h) = -\frac{\pi}{2} \int_{-\infty}^{\infty} h'(u) \tilde{h}(u) du = -\frac{\pi}{2} \int_{-\infty}^{\infty} h'(u) \overline{\tilde{h}(u)} du,$$

where the last equality is valid because h is real-valued. By Parseval's identity ($\|\widehat{f}\|_{L^2} \propto \|f\|_{L^2}$), we then have that

$$Q(h) = -\frac{1}{4} \int_{-\infty}^{\infty} \widehat{h}'(x) \overline{\widehat{h}(x)} dx.$$

Finally, using the well-known identities

$$\widehat{h}'(x) = ix\widehat{h}(x), \quad \widehat{\tilde{h}}(x) = (-i \operatorname{sgn}(x))\widehat{h}(x),$$

it follows that

$$Q(h) = \frac{1}{4} \int_{-\infty}^{\infty} |x| \cdot |\widehat{h}(x)|^2 dx$$

and the claim is immediate. $\square$

Theorem 3.4.1 (Minimization Criterion). *Let $h_0 \in \mathcal{H}$. Assume that for some constant $\lambda \in \mathbb{R}$, the function $p: \mathbb{R} \rightarrow \mathbb{R}$ defined by*

$$p(u) = - \int_{-\infty}^{\infty} h'_0(s) \log|s-u| ds - 2(u \log|u| - u) + \lambda u$$

has the property that, for almost every $u \in \mathbb{R}$,

$$p(u) \text{ is } \begin{cases} = 0 & \text{if } \operatorname{sgn}(u) \cdot h'_0(u) \in (-2, 0); \\ \leq 0 & \text{if } h'_0(u) + \operatorname{sgn}(u) = 1; \\ \geq 0 & \text{if } h'_0(u) + \operatorname{sgn}(u) = -1. \end{cases} \quad (3.4)$$

Then for any $h \in \mathcal{H}$, we have that

$$J(h) \geq J(h_0) + Q(h - h_0).$$

In particular, since $Q(h - h_0) \geq 0$ by Lemma 3.4.1, h_0 is the unique minimizer of J on $\mathcal{H}$.

Proof. Fix $h \in \mathcal{H}$. It is immediate from the property $\operatorname{sgn}(u) \cdot h'(u) \in [-2, 0]$ that $|h'(u) + \operatorname{sgn}(u)| \leq 1$ for all u . The given condition on p then implies that

$$(h'(t) - h'_0(t))p(t) \geq 0$$

for almost every t , since

$$h'(t) - h'_0(t) = h'(t) + \operatorname{sgn}(t) - (h'_0(t) + \operatorname{sgn}(t)).$$

Integrating, we obtain that

$$\int_{-\infty}^{\infty} h'(t)p(t)dt \geq \int_{-\infty}^{\infty} h'_0(t)p(t)dt$$

and by definition of p this translates to

$$2B(h, h_0) + L(h) + \lambda \int_{-\infty}^{\infty} uh'(u)du \geq 2B(h_0, h_0) + L(h_0) + \lambda \int_{-\infty}^{\infty} uh'_0(u)du,$$

where B is a generalized form of the operator Q :

$$B(h_1, h_2) := -\frac{1}{2} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \log |s - t| h'_1(t) h'_2(s) dt ds.$$

Now, because

$$\int_{-\infty}^{\infty} uh'(u)du = -\lim_{a \rightarrow -\infty} \lim_{b \rightarrow \infty} h(u)du = -1$$

by integration by parts, and similarly for h_0 , this furthermore reduces to

$$2B(h - h_0, h_0) \geq L(h_0 - h).$$

Therefore, it follows that

$$\begin{aligned} J(h) &= Q(h) + L(h) + \log 2 = Q((h - h_0) + h_0) + L(h) + \log 2 \\ &= Q(h_0) + Q(h - h_0) + 2B(h_0, h - h_0) + L(h) + \log 2 \\ &\geq Q(h_0) + Q(h - h_0) + L(h_0) + \log 2 \\ &= J(h_0) + Q(h - h_0), \end{aligned}$$

as claimed. $\square$

In light of minimizer proposition, it remains to find an explicit function $h_0 \in \mathcal{H}$ for which the assumptions of this proposition are satisfied. We begin this task by making an informed guess about the form of h_0 . To do this, remember that we arrived at the variational problem by considering the problem of finding the limiting shape of a typical Young diagram chosen according to the Plancherel measure. In particular, recall that

$$\frac{d_\lambda^2}{n!} \sim \exp \left[-n \left(1 + 2I_{\text{hook}}(\phi_\lambda) + O\left(\frac{\log n}{\sqrt{n}}\right) \right) \right].$$

The above relationship implies that the shape for which I_{hook} is minimized is the one most often selected when choosing according to the Plancherel measure. It is therefore natural to suspect h_0 to be of the form $h_0(u) = g_0(u) - |u|$, where g_0 represents the limiting shape expressed in the “Russian” coordinate system we defined earlier:

$$u = \frac{x - y}{\sqrt{2}}, \quad v = \frac{x + y}{\sqrt{2}}.$$

Recall now that by the results of [Section 3.1](#), we also expect uniformly random permutations of order n to have a longest increasing subsequence of length about $2\sqrt{n}$ for large n , and therefore random Young diagrams chosen according to the Plancherel measure to have first row (and, by symmetry, column) length about $2\sqrt{n}$ for large n . Accounting for re-scaling and transformation to the uv coordinate system, this tells us that h_0 should be supported inside the interval $[-\sqrt{2}, \sqrt{2}]$.

Under these assumptions, let us now attempt to use [Theorem 3.4.1](#) to find h_0 . If we impose the first condition in [Eq. \(3.4\)](#), we get that

$$-\int_{-\sqrt{2}}^{\sqrt{2}} h'_0(s) \log |s - u| ds = 2(u \log |u| - u) + \lambda u, \quad |u| \leq \sqrt{2}$$

which upon differentiating with respect to u yields the identity

$$-\int_{-\sqrt{2}}^{\sqrt{2}} \frac{h'_0(s)}{u - s} ds = 2 \log |u| - \lambda.$$

This is useful because the left-hand side determines the Hilbert transform of h'_0 (at least on the interval $[-\sqrt{2}, \sqrt{2}]$), so if we can compute the inverse Hilbert transform of the right-hand side then we should be able to recover h'_0 and subsequently h_0 . This computation is essentially the content of the following Lemma (make the linear change of variables $u = \sqrt{2} \cdot x$).

Lemma 3.4.2. *Consider the function $f: \mathbb{R} \rightarrow \mathbb{R}$ defined by*

$$f(x) = \begin{cases} \arcsin(x) - \frac{\pi}{2} \operatorname{sgn}(x) & \text{if } |x| \leq 1; \\ 0 & \text{if } |x| > 1. \end{cases}$$

Then

$$\tilde{f}(x) = \begin{cases} -\log 2|x| & \text{if } |x| \leq 1; \\ -\log 2|x| - \log(|x| - \sqrt{x^2 - 1}) & \text{if } |x| > 1. \end{cases}$$

We are now in a position where we can fully recover the function h_0 :

Theorem 3.4.2. *The function*

$$h_0(u) = \begin{cases} \frac{2}{\pi} \left(u - \arcsin\left(\frac{u}{\sqrt{2}}\right) + \sqrt{2 - u^2} \right) - |u| & \text{if } |u| \leq \sqrt{2}; \\ 0 & \text{if } |u| > \sqrt{2} \end{cases} \quad (3.5)$$

is the unique minimizer of the functional J on $\mathcal{H}$.

Proof. Before we do anything, we must first check that h_0 is indeed in $\mathcal{H}$. In other words, we must verify that h_0 has the following properties:

1. h_0 is absolutely continuous;
2. $\int_{-\infty}^{\infty} h_0(u) du = 1$;
3. $\operatorname{sgn}(u) \cdot h'_0(u) \in [-2, 0]$ wherever $h'_0(u)$ is defined;
4. h_0 is compactly supported.

The fourth condition is immediate from the definition h_0 . So is the first, in light of the observation that h_0 is continuous and C^1 everywhere except at $u = 0$ (since once can then show that h_0 is Lipschitz continuous, with Lipschitz constant bounded above by $L := \sup_{u \in [-\sqrt{2}, \sqrt{2}] \setminus \{0\}} |h'_0(u)|$, and Lipschitz continuity implies absolute continuity). Integrating by parts, we also easily obtain the second property:

$$\begin{aligned} \int_{-\infty}^{\infty} h_0(u) du &= \int_{-\sqrt{2}}^{\sqrt{2}} h_0(u) du = - \int_{-\sqrt{2}}^{\sqrt{2}} u h'_0(u) du \\ &= \int_{-\sqrt{2}}^{\sqrt{2}} |u| du - \frac{2}{\pi} \int_{-\sqrt{2}}^{\sqrt{2}} u \arcsin \frac{u}{\sqrt{2}} du \\ &= 2 - \frac{4}{\pi} \int_{-1}^1 u \arcsin u du \\ &= 2 - \frac{4}{\pi} \cdot \frac{\pi}{4} = 1. \end{aligned}$$

Finally, for the third property we note that

$$h'_0(u) = \begin{cases} \frac{2}{\pi} \arcsin\left(\frac{u}{\sqrt{2}}\right) - \operatorname{sgn}(u) & \text{if } |u| \leq \sqrt{2}; \\ 0 & \text{if } |u| > \sqrt{2}. \end{cases}$$

Therefore, $\operatorname{sgn}(u) \cdot h'_0(u) = 0$ for $|u| > \sqrt{2}$ and

$$\operatorname{sgn}(u) \cdot h'_0(u) = \frac{2}{\pi} \operatorname{sgn}(u) \arcsin\left(\frac{u}{\sqrt{2}}\right) - 1 \in [-1, 0] \subset [-2, 0]$$

for $|u| \leq \sqrt{2}$ and it follows that h_0 is indeed in $\mathcal{H}$.

We'd now obviously like to complete the proof via an appeal to [Theorem 3.4.1](#). To do this, we must verify that the conditions in [Eq. \(3.4\)](#) are satisfied for a suitable choice of constant λ ; this is exactly where [Lemma 3.4.2](#) comes in handy. Leaving λ as an unknown constant to be determined and differentiating p (as defined in [Theorem 3.4.1](#)), we get

$$p'(u) = - \int_{-\sqrt{2}}^{\sqrt{2}} \frac{h'_0(s)}{u-s} ds - 2 \log |u| + \lambda = -\pi \cdot \widetilde{h'_0}(u) - 2 \log |u| + \lambda.$$

By dilating h_0 so as to fit the conditions of [Lemma 3.4.2](#) (which we can do since the Hilbert transform is a linear operator, and so commutes with linear operations), we find (after a short computation) that

$$p'(u) = \begin{cases} \log 2 + \lambda & \text{if } |u| < \sqrt{2}; \\ \log 2 + \lambda + 2 \log \left(\frac{|u|}{\sqrt{2}} - \sqrt{u^2/2 - 1} \right) & \text{if } |u| > \sqrt{2}. \end{cases}$$

We now claim that the taking $\lambda = -\log 2$ works. Indeed, with this choice of λ the above becomes

$$p'(u) = \begin{cases} 0 & \text{if } |u| < \sqrt{2}; \\ 2 \log \left(\frac{|u|}{\sqrt{2}} - \sqrt{u^2/2 - 1} \right) & \text{if } |u| > \sqrt{2}, \end{cases}$$

whence p must be constant on $[-\sqrt{2}, \sqrt{2}]$. Since p' as defined above is moreover continuous and even, the Fundamental Theorem of Calculus implies that p is odd, so $p(0) = 0$ and p must in fact be identically 0 on $[-\sqrt{2}, \sqrt{2}]$.

Now recall that, by our work above, we know that $\operatorname{sgn}(u) \cdot h'_0(u) \in [-1, 0]$ for all u . Moreover, it is easy to see from our definition of h_0 that $\operatorname{sgn}(u) \cdot h'_0(u) = 0$ for $|u| \geq \sqrt{2}$. This means that when $\operatorname{sgn}(u) \cdot h'_0(u) \in (-2, 0)$, we must have $|u| < \sqrt{2}$ and subsequently that $p(u) = 0$, verifying the first criterion in [Eq. \(3.4\)](#).

Finally, to see that the other two criteria hold, note that $p'(u) \leq 0$ for all u . Therefore, by the Fundamental Theorem of Calculus, we have that

$$p(u) = \int_0^u p'(t) dt \leq 0$$

for $u > \sqrt{2}$ (i.e., when $h'_0(u) + \operatorname{sgn}(u) = 1$), and similarly that

$$p(u) = \int_0^u p'(t) dt = - \int_u^0 p'(t) dt \geq 0$$

for $u < -\sqrt{2}$ (i.e., when $h'_0(u) + \operatorname{sgn}(u) = -1$). This completes the proof. $\square$

We've now found an explicit minimizer for J over $\mathcal{H}$, but we still haven't found what the minimum value actually is. It turns out we do this fairly easily as well:

Theorem 3.4.3. *For h_0 as defined in [Eq. \(3.5\)](#), we have $J(h_0) = -1$.*

Proof. In the proof of the above theorem, we showed that $\int_{-\sqrt{2}}^{\sqrt{2}} u h'_0(u) du = -1$. Therefore,

$$\begin{aligned} L(h_0) &= -2 \int_{-\sqrt{2}}^{\sqrt{2}} h'_0(u) (u \log |u| - u) du \\ &= -2 - 4 \int_0^{\sqrt{2}} h'_0(u) u \log u du \\ &= -2 + 4 \int_0^{\sqrt{2}} u \log u du - \frac{8}{\pi} \int_0^{\sqrt{2}} u \log u \cdot \arcsin \frac{u}{\sqrt{2}} du \\ &= -2 - \log 2. \end{aligned}$$

Next, we note that since $p \equiv 0$ along $[-\sqrt{2}, \sqrt{2}]$, we moreover have that

$$0 = \int_{-\sqrt{2}}^{\sqrt{2}} p(u) h'_0(u) du = 2Q(h_0) + L(h_0) - \lambda = 2Q(h_0) - 2.$$

Therefore, $Q(h_0) = 1$ and it follows that

$$J(h_0) = Q(h_0) + L(h_0) + \log 2 = -1. \quad \square$$

3.5 The Limit Shape Theorem

For each $n \geq 1$, let $\lambda^{(n)}$ denote a random partition of n chosen according to the Plancherel measure of order n . As well, let ϕ_n be the function in the function space $\mathcal{F}$ (recall that $\mathcal{F}$ was our “candidate set” of limit shapes; namely, $\mathcal{F}$ is the collection of all non-increasing compactly supported functions $[0, \infty) \rightarrow [0, \infty)$ which integrate to 1) corresponding to $\lambda^{(n)}$, defined via the relation

$$\phi_n(x) = n^{-1/2}(\lambda^{(n)})'_{\lfloor \sqrt{n}x \rfloor + 1}$$

(i.e., ϕ_n is the outline of the Young diagram $\lambda^{(n)}$ if it were placed in the northeast quadrant of the plane).

It will be natural to formulate our results in the rotated system uv -coordinate system we defined earlier, so let ψ_n denote the element of the function space $\mathcal{G}$ obtained by transforming ϕ_n to this coordinate system. We shall first formulate the limit shape theorem in terms of a non-standard topology on $\mathcal{G}$ and then prove a relation between the metric defining this topology and the uniform norm to obtain the result in the familiar setting.

In particular, recall the functional Q we defined earlier:

$$Q(h) := -\frac{1}{2} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \log |s - t| h'(t) h'(s) dt ds.$$

Proposition 3.5.1. *The functional Q induces a metric d_Q on $\mathcal{G}$ defined by*

$$d_Q(g_1, g_2) = \|g_1 - g_2\|_Q := Q(g_1 - g_2)^{1/2}.$$

Proof. We must verify that d_Q is symmetric, positive-definite, and satisfies the triangle inequality. In illustration of the first, note that for any $g_1, g_2 \in \mathcal{G}$ we have

$$\begin{aligned} Q(g_1 - g_2) &= Q(-(g_2 - g_1)) = -\frac{1}{2} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \log |s - t| [-(g_2 - g_1)]'(t) [-(g_2 - g_1)]'(s) dt ds \\ &= -\frac{1}{2} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \log |s - t| (g_2 - g_1)'(t) (g_2 - g_1)'(s) dt ds \\ &= Q(g_2 - g_1). \end{aligned}$$

Obviously, this implies that $d_Q(g_1, g_2) = d_Q(g_2, g_1)$. For positive-definiteness, we simply observe that

$$d_Q(g_1, g_2) = Q(g_1 - g_2)^{1/2} = Q((g_1 - |u|) - (g_2 - |u|))^{1/2} \geq 0$$

by [Lemma 3.4.1](#) (since every $g \in \mathcal{G}$ is Lipschitz and has the property that $g(u) - |u|$ is compactly supported), with equality if and only if $(g_1 - |u|) - (g_2 - |u|) \equiv 0$, i.e. if and only if $g_1 = g_2$. Finally, for the triangle inequality, note that if $g_3 \in \mathcal{G}$ is any third function, then

$$d_Q(g_1, g_2) = Q((g_1 - g_3) - (g_2 - g_3))^{1/2} = [Q(g_1 - g_3) - 2B(g_1 - g_3, g_2 - g_3) + Q(g_2 - g_3)]^{1/2}.$$

If we assume that the above is less than or equal to $Q(g_1 - g_3)^{1/2} + Q(g_2 - g_3)^{1/2}$, then by squaring both sides and simplifying we find that the triangle inequality holds if and only if

$$-B(g_1 - g_3, g_2 - g_3) \leq Q(g_1 - g_3)^{1/2} Q(g_2 - g_3)^{1/2}, \quad (3.6)$$

By replacing $g_1 - g_3$ with $(g_1 - |u|) - (g_3 - |u|)$ and similarly for $g_2 - g_3$, it shall suffice to show that the above holds for compactly supported Lipschitz functions $f, g: \mathbb{R} \rightarrow \mathbb{R}$ which integrate to 0. To that end, we first note that because f and g are Lipschitz and compactly supported, their Hilbert transforms exist everywhere. Moreover, by integrating by parts we derive the identity

$$\tilde{g}(s) = \frac{1}{\pi} \int_{-\infty}^{\infty} g'(t) \log |s - t| dt.$$

and similarly for f . With this, [Eq. \(3.6\)](#) becomes

$$\int_{-\infty}^{\infty} f'(s) \tilde{g}(s) ds \leq \sqrt{\int_{-\infty}^{\infty} f'(s) \tilde{f}(s) ds} \cdot \sqrt{\int_{-\infty}^{\infty} g'(s) \tilde{g}(s) ds}$$

Now note that because

$$\int_{-\infty}^{\infty} f'(s) \widehat{g}(s) ds = \langle f', \widehat{g} \rangle_{L^2} = \langle \widehat{f}', \widehat{g} \rangle_{L^2} = 2\pi \int_{-\infty}^{\infty} |s| \cdot \widehat{f}(s) \widehat{g}(s) ds,$$

(where to obtain the last equality we used the identities $\widehat{f}'(x) = 2\pi i x \cdot \widehat{f}(x)$ and $\widehat{\widehat{g}}(x) = -i \operatorname{sgn}(x) \cdot \widehat{g}(x)$), we can further simplify Eq. (3.6) to

$$\int_{-\infty}^{\infty} |s| \cdot \widehat{f}(s) \widehat{g}(s) ds \leq \sqrt{\int_{-\infty}^{\infty} |s| \cdot \widehat{f}(s)^2 ds} \cdot \sqrt{\int_{-\infty}^{\infty} |s| \cdot \widehat{g}(s)^2 ds}.$$

To show this holds, it shall suffice (by the Cauchy-Schwarz inequality) to show that $\langle f, g \rangle := \int_{-\infty}^{\infty} |s| \cdot \widehat{f}(s) \widehat{g}(s) ds$ defines an inner product on the space of compactly supported Lipschitz functions $\mathbb{R} \rightarrow \mathbb{R}$ which integrate to 0. This, however, is clear. $\square$

Theorem 3.5.1. *Under the metric d_Q , the random function ψ_n converges in probability to the limiting shape given by*

$$\Omega(u) = \begin{cases} \frac{2}{\pi} \left(u \arcsin \frac{u}{\sqrt{2}} + \sqrt{2 - u^2} \right) & \text{if } u \leq \sqrt{2}; \\ |u| & \text{if } |u| > \sqrt{2}. \end{cases}$$

In other words, for every $\varepsilon > 0$ we have that

$$\mathbb{P}(\|\psi_n - \Omega\|_Q > \varepsilon) \rightarrow 0.$$

Proof. For each partition λ of n , denote by g_λ the element of $\mathcal{G}$ corresponding to the function $\phi_\lambda \in \mathcal{F}$, and let $h_\lambda(u) = g_\lambda(u) - |u|$. Moreover, fix $\varepsilon > 0$ and denote by $\mathcal{M}_n$ the set of partitions λ of n for which $\|g_\lambda - \Omega\|_Q > \varepsilon$. By Theorem 3.4.2, we have, for each $\lambda \in \mathcal{M}_n$,

$$J(h_\lambda) \geq J(h_0) + Q(h_\lambda - h_0) = -1 + Q(g_\lambda - \Omega) > -1 + \varepsilon^2.$$

Therefore, by {correspondence} it follows that

$$\mathbb{P}(\lambda^{(n)} = \lambda) = \exp \left[-n \left(1 + J(h_\lambda) + O \left(\frac{\log n}{\sqrt{n}} \right) \right) \right] \leq \exp \left[-\varepsilon^2 n + O(\sqrt{n} \log n) \right].$$

Finally, we note that because (letting $\mathcal{P}(n)$ denote the set of all partitions of n) $|\mathcal{M}_n| \leq |\mathcal{P}(n)|$, and $|\mathcal{P}(n)| \leq e^{C\sqrt{n}}$ by a famous result of Hardy and Ramanujan, it follows that

$$\mathbb{P}(\|\psi_n - \Omega\|_Q > \varepsilon) = \mathbb{P}(\lambda^{(n)} \in \mathcal{M}_n) = \sum_{\lambda \in \mathcal{M}_n} \mathbb{P}(\lambda^{(n)} = \lambda) \leq C \exp(-\varepsilon^2 n + C\sqrt{n} + O(\sqrt{n} \log n)),$$

which converges to 0 as $n \rightarrow \infty$. $\square$

We now give a Lemma which shall relate the exotic $\|\cdot\|_Q$ norm and the familiar supremum norm to get the limit shape result that we truly desire.

Lemma 3.5.1. *Given $A, L > 0$, let $\operatorname{Lip}(L, A)$ denote the space of all L -Lipschitz functions $f: \mathbb{R} \rightarrow \mathbb{R}$ which are supported in the interval $[-A, A]$. Then for any $f \in \operatorname{Lip}(L, A)$, we have that*

$$\|f\|_u = \|f\|_{L^\infty} \leq C Q(f)^{1/4},$$

where $C > 0$ is a constant that depends on A and L .

Proof. Recall that, by the argument of Lemma 3.4.1, the action of Q on any given $f \in \operatorname{Lip}(L, A)$ can be written as

$$Q(f) = \frac{1}{4} \int_{-\infty}^{\infty} |x| \cdot |\widehat{f}(x)|^2 dx = \frac{1}{4} \int_{-\infty}^{\infty} |x|^{1/2} \widehat{f}(x)^2 dx.$$

If we further recall Plancherel's theorem ($\|\widehat{f}\|_{L^2} = \|f\|_{L^2}$) from Fourier analysis, we see that $Q(f)^{1/2}$ is proportional to the L^2 -norm of a function g_f whose Fourier transform has the property that $|\widehat{g}_f(\xi)| = |\xi|^{1/2}|\widehat{f}(\xi)|$. In fact, we claim that

$$g_f(x) = \frac{1}{2\sqrt{\pi}} \int_0^\infty \frac{f(x) - f(x-t)}{t^{3/2}} dt = \frac{1}{2\sqrt{\pi}} \int_{-\infty}^x \frac{f(x) - f(y)}{(x-y)^{3/2}} dy$$

is such a function, and we shall call g_f the *half-derivative* or *derivative of order 1/2* of f .

To see that g_f is well-defined, note that because f is supported on $[-A, A]$, we have that

$$g_f(x) = \begin{cases} 0 & \text{if } x \leq -A; \\ \frac{f(x)}{\sqrt{\pi(x+A)}} + \frac{1}{2\sqrt{\pi}} \int_{-A}^x \frac{f(x)-f(y)}{(x-y)^{3/2}} dy & \text{if } -A < x < A; \\ -\frac{1}{2\sqrt{\pi}} \int_{-A}^A \frac{f(y)}{(x-y)^{3/2}} dy & \text{if } x \geq A. \end{cases}$$

The first and third cases should be clear enough. The second case is obtained by breaking the domain of integration into $(-\infty, -A] \cup [-A, x]$ and using the fact that $f(y) = 0$ along the former; this will yield a convergent integral that simplifies to the first term. Moreover, the Lipschitz property of f guarantees that the second term converges absolutely.

Next we note that with this presentation of g_f , it is easy to see that g_f is bounded and for large positive x satisfies $|g_f(x)| \leq Cx^{-3/2}$ for constant $C > 0$. In particular, $g_f \in L^1$ and so has a Fourier transform. This Fourier transform can be computed explicitly as follows:

$$\begin{aligned} \widehat{g}_f(\xi) &= \int_{-\infty}^\infty g_f(x) e^{-i\xi x} dx = \frac{1}{2\sqrt{\pi}} \int_0^\infty \frac{1}{t^{3/2}} \int_{-\infty}^\infty (f(x) - f(x-t)) e^{-i\xi x} dx dt \\ &= \frac{1}{2\sqrt{\pi}} \int_0^\infty \frac{1}{t^{3/2}} (\widehat{f}(\xi) - \widehat{f}(\xi) e^{-i\xi t}) dt \\ &= \frac{1}{2\sqrt{\pi}} \widehat{f}(\xi) \int_0^\infty \frac{1 - e^{-i\xi t}}{t^{3/2}} dt \\ &= e^{-\pi i \operatorname{sgn}(\xi)/4} |\xi|^{1/2} \widehat{f}(\xi). \end{aligned}$$

This shows that g_f indeed has the property that $|\widehat{g}_f(\xi)| = |\xi|^{1/2}|\widehat{f}(\xi)|$.

At this point, I'd like to extend my appreciation to the reader for going through of all this without any idea as to why we should care about the function g_f . The reason is because our goal is to relate $\|f\|_\infty$ to $Q(f)$, and the defining property of g_f is such that we can achieve this by instead relating $\|f\|_\infty$ to $\|g_f\|_2$. It should moreover be simpler to do this because we ought to be able to recover f from g_f (the half-derivative of f) by performing a “fractional integration of order 1/2.” In particular, consider the function h_f defined by

$$h_f(x) = \frac{1}{\sqrt{\pi}} \int_{-\infty}^x \frac{g_f(t)}{\sqrt{x-t}} dt = \begin{cases} 0 & \text{if } x < -A; \\ \frac{1}{\sqrt{\pi}} \int_{-A}^x \frac{g_f(t)}{\sqrt{x-t}} dt & \text{if } x \geq -A. \end{cases}$$

Then for all $\xi \neq 0$ we have that

$$\begin{aligned} \widehat{h}_f(\xi) &= \int_{-A}^\infty h_f(x) e^{-i\xi x} dx = \frac{1}{\sqrt{\pi}} \int_{-A}^\infty e^{-i\xi x} \int_{-A}^x \frac{g_f(t)}{\sqrt{x-t}} dt dx \\ &= \frac{1}{\sqrt{\pi}} \int_{-A}^\infty e^{-i\xi x} \int_{-A}^\infty \chi_{\{t \leq x\}} \cdot \frac{g_f(t)}{\sqrt{x-t}} dt dx \\ &= \frac{1}{\sqrt{\pi}} \int_{-A}^\infty g_f(t) \int_t^\infty \frac{e^{-i\xi x}}{\sqrt{x-t}} dx dt \\ &= \frac{1}{\sqrt{\pi}} \int_{-A}^\infty g_f(t) e^{-i\xi t} dt \int_0^\infty \frac{e^{-i\xi u}}{\sqrt{u}} du \\ &= e^{\pi i \operatorname{sgn}(\xi)/4} |\xi|^{-1/2} \widehat{g}_f(\xi) \\ &= \widehat{f}(\xi). \end{aligned}$$

Therefore, $f = h_f$ almost everywhere and so

$$\begin{aligned}
\|f\|_{L^1} &= \int_{-A}^A |f(x)| dx \leq \frac{1}{\sqrt{\pi}} \int_{-A}^A \int_{-A}^x \frac{|g_f(t)|}{\sqrt{x-t}} dt dx \\
&= \frac{1}{\sqrt{\pi}} \int_{-A}^A |g_f(t)| \left(\int_t^A \frac{1}{\sqrt{x-t}} dx \right) dt \\
&= \frac{1}{\sqrt{\pi}} \int_{-A}^A |g_f(t)| 2\sqrt{A-t} dt \\
&\leq \frac{2\sqrt{2A}}{\sqrt{\pi}} \int_{-A}^A |g_f(t)| dt \\
&= \frac{2\sqrt{2A}}{\sqrt{\pi}} \|g_f \cdot \chi_{[-A,A]}\|_{L^1} \\
&\leq \frac{4A}{\sqrt{\pi}} \|g_f\|_{L^2} \\
&= \frac{4\sqrt{2A}}{\pi} Q(f)^{1/2},
\end{aligned}$$

where the last equality follows from the fact that $\|g_f\|_{L^2}^2 = \|\widehat{g_f}\|_{L^2}^2/2\pi = (2/\pi)Q(f)$. Finally, if we take $x_0 \in [-A, A]$ such that $|f(x_0)| = \|f\|_{L^\infty} = \|f\|_u$, then

$$\begin{aligned}
\|f\|_{L^1} &= \int_{-\infty}^{\infty} |f(x)| dx \geq \int_{-\infty}^{\infty} \max\{0, |f(x_0)| - L|x - x_0|\} dx \\
&= \int_{x_0 - |f(x_0)|/L}^{x_0 + |f(x_0)|/L} |f(x_0)| - L|x - x_0| dx \\
&= 2 \frac{|f(x_0)|^2}{L} - L \int_{-|f(x_0)|/L}^{|f(x_0)|/L} |u| du \\
&= 2 \frac{|f(x_0)|^2}{L} - \frac{|f(x_0)|^2}{L} \\
&= \|f\|_u^2 / L.
\end{aligned}$$

Combining this with the previous bound we found for $\|f\|_{L^1}$, we obtain that

$$\|f\|_u^2 \leq \frac{4\sqrt{2AL}}{\pi} Q(f)^{1/2}$$

and the claim follows. $\square$

With this Lemma, we are finally ready to prove the following famous 1977 result of Vershik, Kerov, Logan, and Shepp:

Theorem 3.5.2 (Limit Shape Theorem for Plancherel-Random Partitions). *With respect to the uniform norm, the random function ψ_n converges in probability to the limiting shape Ω defined in [Theorem 3.5.1](#).*

Proof. We first note that by [Corollary 3.1.2](#), the probability that the random partition $\lambda^{(n)}$ has first row/column length less than $3\sqrt{n}$ converges to 1 as $n \rightarrow \infty$. This means that, with probability converging to 1, $\rho_n := \psi_n - |u|$ is supported on the interval $[-3\sqrt{2}, 3\sqrt{2}]$. If we let $C > 0$ denote the constant associated to the space of 1-Lipschitz functions supported on this interval as per [Lemma 3.5.1](#) (and note that we may indeed apply this Lemma to $\psi_n - \Omega$ since $\psi_n - \Omega = (\psi_n - |u|) - h_0$ is always compactly supported 1-Lipschitz function) then, given $\varepsilon > 0$,

$$\begin{aligned}
\mathbb{P}(\|\psi_n - \Omega\|_u > \varepsilon) &= \mathbb{P}(\|\psi_n - \Omega\|_u > \varepsilon \text{ and } \text{supp } \rho_n \subseteq [-3\sqrt{2}, 3\sqrt{2}]) \\
&\quad + \mathbb{P}(\|\psi_n - \Omega\|_u > \varepsilon \text{ and } \text{supp } \rho_n \not\subseteq [-3\sqrt{2}, 3\sqrt{2}]) \\
&\leq \mathbb{P}(C\|\psi_n - \Omega\|_Q^{1/2} > \varepsilon) + \mathbb{P}(\text{supp } \rho_n \not\subseteq [-3\sqrt{2}, 3\sqrt{2}]).
\end{aligned}$$

Since both terms on the right-hand side converge to 0 by [Theorem 3.5.1](#) and [Corollary 3.1.2](#), the result follows. $\square$

Chapter 4

The Dimer Model

Given a graph G , a **perfect matching** or **dimer configuration** of G is a choice of edges of G such that every vertex of G is adjacent to exactly one of these edges. Roughly speaking, the **dimer model** is concerned with the study of the set of dimer configurations of certain graphs. For our purposes, let us assume that G is a finite planar graph with edge set E and vertex set V . The goal of this chapter will be to provide an exact solution to the following problem: how many dimer configurations of G are there?

4.1 The Pfaffian

Definition 4.1.1. Let $A = (a_{ij})$ be a skew-symmetric $2n \times 2n$ matrix. We define the **Pfaffian** of A to be the solution $\text{pf}(A)$ to the equation

$$\omega^{\wedge n} = 2^n n! \text{pf}(A) e_1 \wedge \cdots \wedge e_{2n},$$

where $(e_1, \dots, e_{2n})$ denotes the standard basis of $\mathbb{R}^{2n}$ and $\omega = \sum_{i,j} a_{ij} e_i \wedge e_j$. By carrying out the calculation $\omega^{\wedge n}$, one finds that

$$\text{pf}(A) = \frac{1}{2^n n!} \sum_{\sigma \in S_{2n}} \text{sgn}(\sigma) a_{\sigma(1)\sigma(2)} \cdots a_{\sigma(2n-1)\sigma(2n)}.$$

Proposition 4.1.1. Again let $A = (a_{ij})$ be a skew-symmetric $2n \times 2n$ matrix. Then, for any $2n \times 2n$ matrix $B = (b_{ij})$, we have that $\text{pf}(BAB^\top) = \det(B) \text{pf}(A)$.

Proof. Note that

$$BAB^\top \mapsto \sum_{i,j} \left(\sum_{k,l} b_{ik} a_{kl} b_{jl} \right) e_i \wedge e_j = \sum_{k,l} a_{kl} f_k \wedge f_l,$$

where $f_k = \sum_i b_{ik} e_i$. Therefore,

$$2^n n! \text{pf}(BAB^\top) e_1 \wedge \cdots \wedge e_{2n} = \omega^{\wedge n} = 2^n n! \text{pf}(A) f_1 \wedge \cdots \wedge f_{2n} = 2^n n! \det(B) \text{pf}(A) e_1 \wedge \cdots \wedge e_{2n},$$

and it follows that $\text{pf}(BAB^\top) = \det(B) \text{pf}(A)$, as desired. $\square$

Proposition 4.1.2. The value of $\text{sgn}(\sigma) a_{\sigma(1)\sigma(2)} \cdots a_{\sigma(2n-1)\sigma(2n)}$ is entirely determined by the “matching” of $\{1, \dots, 2n\}$ into unordered pairs $\{\{\sigma(1), \sigma(2)\}, \dots, \{\sigma(2n-1), \sigma(2n)\}\}$. In particular, if we let

$$\mathcal{M}_{2n} = \{\pi = \{\{i_1, j_1\}, \dots, \{i_n, j_n\}\}$$

denote the set of matchings of $\{1, \dots, 2n\}$ into unordered pairs, then since there are $2^n n!$ permutations which induce any given matching ($n!$ ways to assign each pair $\{\sigma(k), \sigma(k+1)\}$ to a pair $\{i_\ell, j_\ell\}$ in the matching, plus an additional factor of 2^n since it doesn't matter which of $\sigma(k)$ or $\sigma(k+1)$ gets assigned to which number in the pair) we may write

$$\text{pf}(A) = \sum_{\pi \in \mathcal{M}_{2n}} \text{sgn} \begin{pmatrix} 1 & 2 & \cdots & 2n-1 & 2n \\ i_1 & j_1 & \cdots & i_n & j_n \end{pmatrix} a_{i_1 j_1} \cdots a_{i_n j_n}.$$

Proof. Suppose that $\sigma, \tau \in S_{2n}$ are two (distinct) permutations which induce the same matching in $\mathcal{M}_{2n}$. We must show that

$$\operatorname{sgn}(\sigma)a_{\sigma(1)\sigma(2)} \cdots a_{\sigma(2n-1)\sigma(2n)} = \operatorname{sgn}(\tau)a_{\tau(1)\tau(2)} \cdots a_{\tau(2n-1)\tau(2n)}$$

We proceed by induction on the order $2n$ of the permutations. If $n = 1$, then without loss of generality $\sigma = \operatorname{id}$ and $\tau = \tau_{12}$, the transposition $1 \leftrightarrow 2$. By skew-symmetry,

$$\operatorname{sgn}(\sigma)a_{\sigma(1)\sigma(2)} = a_{12} = -a_{21} = \operatorname{sgn}(\tau)a_{\tau(1)\tau(2)}$$

and the base case is proven. By induction, suppose now that the claim is true for all permutations of order $2n$, and that $\sigma, \tau \in S_{2n+2}$ are such that $[\sigma] = [\tau]$ in $\mathcal{M}_{2n+2}$. Without loss of generality, let us assume that this matching is $[\operatorname{id}] = \{\{1, 2\}, \dots, \{2n+1, 2n+2\}\}$. If σ' is let to denote the permutation which interchanges $2n+1$ and $\sigma(2n+1)$, interchanges $2n+2$ and $\sigma(2n+2)$, and fixes everything else, then the permutation $\sigma'' := \sigma'\sigma$ fixes $2n+1$ and $2n+2$ and thus may be regarded as a permutation of order $2n$. Define $\tau'' := \tau'\tau$ similarly. Since $\operatorname{sgn}(\sigma') = \operatorname{sgn}(\tau') = 1$, the induction hypothesis implies that

$$\operatorname{sgn}(\sigma)a_{\sigma''(1)\sigma''(2)} \cdots a_{(2n+1)(2n+2)} = \operatorname{sgn}(\tau)a_{\tau''(1)\tau''(2)} \cdots a_{(2n+1)(2n+2)}.$$

On other hand, for each $i = 1, \dots, n+1$, we have that $a_{\sigma''(2i-1)\sigma''(2i)} = a_{\sigma(2i-1)\sigma(2i)}$ unless $\{2i-1, 2i\} \cap \{2n+1, 2n+2, \sigma^{-1}(2n+1), \sigma^{-1}(2n+2)\} \neq \emptyset$. If without loss of generality $\{\sigma^{-1}(2n+1), \sigma^{-1}(2n+2)\} = \{1, 2\}$, then by skew-symmetry

$$\operatorname{sgn}(\sigma)a_{\sigma''(1)\sigma''(2)} \cdots a_{(2n+1)(2n+2)} = \operatorname{sgn}(\sigma)a_{\sigma(2n+1)\sigma(2n+2)}a_{\sigma(3)\sigma(4)} \cdots a_{\sigma(1)\sigma(2)}.$$

By the same argument $\operatorname{sgn}(\tau)a_{\tau''(1)\tau''(2)} \cdots a_{(2n+1)(2n+2)} = \operatorname{sgn}(\tau)a_{\tau(1)\tau(2)} \cdots a_{\tau(2n+1)\tau(2n+2)}$, so the claim follows. $\square$

Corollary 4.1.1. *For any skew-symmetric $2n \times 2n$ matrix A , we have that $\operatorname{pf}(A)^2 = \det(A)$.*

Proof. Recall that by the spectral theory of skew-symmetric real matrices we may write $A = QBQ^\top$, where Q is orthogonal and B is block diagonal with blocks B_i of the form

$$B_i = \begin{pmatrix} 0 & \lambda_i \\ -\lambda_i & 0 \end{pmatrix}.$$

By [Proposition 4.1.1](#), then, we have that $\operatorname{pf}(A)^2 = \det(Q)^2 \operatorname{pf}(B)^2 = \operatorname{pf}(B)^2$ and it shall suffice (since $\det(B) = \det(A)$) to show that $\operatorname{pf}(B)^2 = \det(B)$. In illustration of this, we note that the only nonzero entries of B are $b_{2m-1, 2m}$ and $b_{2m, 2m-1}$ for $m = 1, \dots, n$. Therefore, the only permutations $\sigma \in S_{2n}$ which contribute to the Pfaffian of B are those whose corresponding matching of $\{1, \dots, 2n\}$ into unordered pairs $\{\{\sigma(1), \sigma(2)\}, \dots, \{\sigma(2n-1), \sigma(2n)\}\}$ consists precisely of the pairs $\{2m-1, 2m\}$. By [Proposition 4.1.2](#), then, we have that

$$\operatorname{pf}(B) = \operatorname{sgn} \begin{pmatrix} 1 & 2 & \cdots & 2n-1 & 2n \\ 1 & 2 & \cdots & 2n-1 & 2n \end{pmatrix} a_{12} \cdots a_{(2n-1)(2n)} = \prod_i \lambda_i$$

and it follows that

$$\operatorname{pf}(B)^2 = \prod_i \lambda_i^2 = \prod_i \det(B_i) = \det(B),$$

as desired. $\square$

4.2 The FKT Algorithm

Recall that we are working over a fixed finite planar graph G . In this section we will see how to count the number of perfect matchings of G (and, in fact, any weighted sum of the dimer configurations of G). To avoid the trivial case, let us assume that G admits at least one perfect matching; obviously, this requires that G have an even number of vertices.

Let us define an **edge-weight system** on G to be any positive real-valued function ν on the set of edges of G . Note that each such function ν determines a probability measure on the set of dimer configurations $\mathcal{D}$ of G via

$$\mathbb{P}(D) = \frac{\nu(D)}{Z(G; \nu)},$$

where $\nu(D) = \prod_{e \in D} \nu(e)$ and $Z(G; \nu) = \sum_{D \in \mathcal{D}} \nu(D)$. Here, we give the normalization constant $Z(G; \nu)$ a special name: the *partition function* for the dimer model on G with weight system ν . Note that if the weight system is constant equal to 1, then $Z(G; \nu)$ is simply the number of dimer configurations of G . The goal of the Fisher-Kasteleyn-Temperley (FKT) is to express the dimer partition function of G as the Pfaffian of a certain matrix. This is desirable because, as per [Corollary 4.1.1](#), the Pfaffian of a matrix can be efficiently computed via standard determinant algorithms. It proceeds as follows:

Enumerate the vertices of G by $1, 2, \dots, 2n$ and fix an orientation K of the edges of G (by this, we mean a choice of orientation for each edge of G , and not a single orientation that should be the same across all edges of G). Let $A = (a_{ij})$ denote the associated weighted adjacency matrix; that is, the skew-symmetric $2n \times 2n$ matrix whose ij entry is defined by

$$a_{ij} = \sum_{e: i \leftrightarrow j} \varepsilon_{ij}(e) \nu(e),$$

where the summation extends over all edges $e \in E$ between i and j , and

$$\varepsilon_{ij}(e) = \begin{cases} 1 & \text{if } e: i \rightarrow j; \\ -1 & \text{if } e: j \rightarrow i; \end{cases}$$

that is, 1 if e is oriented from i to j and -1 if conversely e is oriented from j to i .

Now recall that by [Proposition 4.1.2](#),

$$\text{pf}(A) = \sum_{\pi \in \mathcal{M}_{2n}} \text{sgn} \begin{pmatrix} 1 & 2 & \cdots & 2n-1 & 2n \\ i_1 & j_1 & \cdots & i_n & j_n \end{pmatrix} a_{i_1 j_1} \cdots a_{i_n j_n}.$$

Notice that to each matching $\pi = \{\{i_1, j_1\}, \dots, \{i_n, j_n\}\}$, there are $\prod_{\ell=1}^n (\# \text{ of edges between } i_\ell \text{ and } j_\ell)$ corresponding dimer configurations; namely, each $\{e_1, \dots, e_n\}$ where e_ℓ is any choice of edge between i_ℓ and j_ℓ . Let us denote the set of such dimer configurations by $\mathcal{D}_\pi$. Then by definition of A , the above reduces to

$$\begin{aligned} \text{pf}(A) &= \sum_{\pi \in \mathcal{M}_{2n}} \text{sgn} \begin{pmatrix} 1 & 2 & \cdots & 2n-1 & 2n \\ i_1 & j_1 & \cdots & i_n & j_n \end{pmatrix} \prod_{\ell=1}^n \sum_{e: i_\ell \leftrightarrow j_\ell} \varepsilon_{i_\ell j_\ell}(e) \nu(e) \\ &= \sum_{\pi \in \mathcal{M}_{2n}} \text{sgn} \begin{pmatrix} 1 & 2 & \cdots & 2n-1 & 2n \\ i_1 & j_1 & \cdots & i_n & j_n \end{pmatrix} \sum_{D \in \mathcal{D}_\pi} \prod_{\ell=1}^n \varepsilon_{i_\ell j_\ell}(e_\ell) \nu(e_\ell) \\ &= \sum_{\pi \in \mathcal{M}_{2n}} \sum_{D \in \mathcal{D}_\pi} \left[\text{sgn} \begin{pmatrix} 1 & 2 & \cdots & 2n-1 & 2n \\ i_1 & j_1 & \cdots & i_n & j_n \end{pmatrix} \prod_{\ell=1}^n \varepsilon_{i_\ell j_\ell}(e_\ell) \right] \prod_{\ell=1}^n \nu(e_\ell) \\ &= \sum_{D \in \mathcal{D}} \varepsilon(D) \nu(D), \end{aligned}$$

where $\varepsilon(D) = \text{sgn}(\sigma_D) \prod_{\ell=1}^n \varepsilon_{i_\ell j_\ell}(e_\ell) \in \{\pm 1\}$, $\sigma_D: (1, \dots, 2n) \mapsto (i_1, j_1, \dots, i_n, j_n)$. We see from this that if we can choose our orientation K of the edges of G so that the signs $\varepsilon(D)$ of each summand are aligned, then it will be the case that $Z(G; \nu) = |\text{pf}(A)|$. Such an orientation shall be called a *Pfaffian orientation*, and was proven by Kasteleyn to always exist for planar graphs.

Theorem 4.2.1 (Kasteleyn's Theorem). *There exists a Pfaffian orientation K of the edges of G .*

We break the proof of this theorem up into several Lemmas.

Lemma 4.2.1. *Given distinct dimer configurations $D, D' \in \mathcal{D}$ of G , the symmetric difference $D \Delta D'$ consists of the disjoint union of simple closed curves of even length in G , the edges comprising which alternate between D and D' .*

Proof. Consider an edge e_0 in the symmetric difference $D\Delta D'$, with endpoints v_0^1 and v_0^2 . Because D' is a perfect matching, there exists a unique edge $e'_0 \in D'$ with v_0^1 as an endpoint. Moreover, because e_0 lies in the symmetric difference, e'_0 must be distinct from e_0 . Let $(v_0^2)'$ denote the other endpoint of e'_0 . If $(v_0^2)' = v_0^2$, then we are done. If, on the other hand, $(v_0^2)'$ is distinct from v_0^2 , then because D is a perfect matching there is a unique edge $e_1 \in D$ with $v_1^1 := (v_0^2)'$ as an endpoint. In this case, perfection of D also guarantees that the other endpoint v_1^2 of e_1 must be distinct from v_0^2 . Thus, let us use perfection of D' to find $e'_1 \in D'$ having $(v_1^1)' := v_1^1$ as one of its endpoints and $(v_1^2)'$ as the other. If $(v_1^2)' = v_0^2$, we are done. If not, we simply continue this process; by finiteness of G it must terminate after some number of steps m , whence the edges $e_0, e'_0, e_1, e'_1, \dots, e_m, e'_m$ form a simple closed curve of even length, the edges of which alternate between D and D' . It follows that $D\Delta D'$ is the union of such curves; that this union is disjoint is immediate from the fact that D and D' are perfect matchings. $\square$

Note that to prove [Theorem 4.2.1](#), it shall suffice to show that $\varepsilon(D)\varepsilon(D') = 1$ for any given (distinct) dimer configurations $D, D' \in \mathcal{D}$. Let $C_1, \dots, C_m$ denote the curves comprising the symmetric difference $D\Delta D'$, as per the above lemma. Each such curve corresponds to a cyclic permutation σ_{C_ℓ} in S_{2n} ; namely, the cycle which traverses the curve C_ℓ clockwise (the choice of clockwise might seem important here, but we shall see later that it doesn't matter, and we could have equally well picked counter-clockwise; also note that we fixed an enumeration of the vertices of G earlier, so what we really mean here is the cyclic permutation which, for each edge in C_ℓ , maps the number associated with the first endpoint to the number associated with the other endpoint, according to this clockwise orientation). Because the length of C_ℓ is even, $\text{sgn}(\sigma_{C_\ell}) = -1$. Consequently, the permutation $\tau := \prod_{\ell=1}^m \sigma_{C_\ell}$ has sign $(-1)^m$.

Now recall that by our work above, there are permutations $\sigma_D, \sigma_{D'} \in S_{2n}$ associated to D and D' that arise in $\varepsilon(D)$ and $\varepsilon(D')$. In fact, by [Proposition 4.1.2](#) there are many such permutations, and the specific choice of representative we used earlier is unimportant and may be changed.

Lemma 4.2.2. *It is possible to choose representatives so that $\sigma_{D'} \circ \sigma_D^{-1} = \tau$.*

Proof. For clarity, let us suppose that D is given by edges $e_1, \dots, e_n$ matching vertices i_ℓ to j_ℓ , and that D' is given by edges $e'_1, \dots, e'_n$ matching vertices i'_ℓ to j'_ℓ (notice the use of word *to* here, which is to reflect our convention of orienting the curves comprising $D\Delta D'$ clockwise). By [Proposition 4.1.2](#), any permutation which sends each pair $\{k, k+1\}$ to a pair $\{i_\ell, j_\ell\}$ is a valid representative of D , and likewise for D' . Consider the representative $\sigma_D: (1, \dots, 2n) \mapsto (i_1, j_1, \dots, i_n, j_n)$ of D , so that $\sigma_D^{-1}: (i_1, j_1, \dots, i_n, j_n) \mapsto (1, 2, \dots, 2n-1, 2n)$. Let us now construct a suitable representative of D' . Consider the edge in D joining i_1 and j_1 . If this same edge occurs in D' , then let $\sigma_{D'}$ map $(1, 2)$ to the pair (i_1, j_1) . Conversely, if this edge lies exclusively in D (meaning it also lies in the symmetric difference $D\Delta D'$, say on the curve C_1 as per [Lemma 4.2.1](#)). Then by [Lemma 4.2.1](#) $j_1 = i'_\ell$ for some ℓ . In this case, let $\sigma_{D'}$ map the pair $(1, 2)$ to the pair (i'_ℓ, j'_ℓ) . Continue matching pairs in this way until $\sigma_{D'} \circ \sigma_D^{-1}$ traverses C_1 in the clockwise fashion. If doing so defined $\sigma_{D'}$ completely, then we are done. Otherwise, there are pairs $(k, k+1), \dots, (2n-1, 2n)$ left that still need to be assigned to something. In this case, simply rerun the proof, but with $(1, 2)$ replaced by $(k, k+1)$. Do this until all curves $C_1, \dots, C_m$ in the symmetric difference $D\Delta D'$ are accounted for. It is not hard to see that the resulting representative $\sigma_{D'}$ satisfies $\sigma_{D'} \circ \sigma_D = \tau$. $\square$

Using this choice of representatives, we have (since $\text{sgn}(\sigma_{D'})\text{sgn}(\sigma_D) = \text{sgn}(\sigma_{D'})\text{sgn}(\sigma_D^{-1}) = \text{sgn}(\sigma_{D'} \circ \sigma_D^{-1}) = \text{sgn}(\tau) = (-1)^m$)

$$\varepsilon(D)\varepsilon(D') = (-1)^m \prod_{\ell=1}^n \varepsilon_{i_\ell j_\ell}(e_\ell) \prod_{k=1}^n \varepsilon_{i'_k j'_k}(e'_k).$$

Now note that if an edge occurs in both D and D' , then we can pair off their corresponding ε terms in the above products to get a factor of 1. The products thus reduce to the product of the ε terms corresponding to the edges in the symmetric difference $D\Delta D'$. If we then again re-arrange the terms so that the ε terms for the edges in each curve C_ℓ (as per [Lemma 4.2.1](#)) are together, then we get

$$\varepsilon(D)\varepsilon(D') = (-1)^m \prod_{\ell=1}^m \text{sgn}(C_\ell), \quad \text{sgn}(C_\ell) = \prod_{\text{edges in } C_\ell \cap D} \varepsilon \text{ term} \prod_{\text{edges in } C_\ell \cap D'} \varepsilon \text{ term}.$$

Recall now that the clockwise orientation on C_ℓ is such that, for each edge e_k joining i_k and j_k therein, i_k is oriented towards j_k . It follows that if the orientation e_k inherits from C_ℓ is the same as that which it inherits from the chosen orientation K of G , then $\varepsilon_{i_k j_k}(e_k) = 1$. In other words, we have $\text{sgn}(C_\ell) = (-1)^{n(C_\ell)}$, where $n(C_\ell)$ denotes the number of edges in C_ℓ whose true orientation (i.e orientation as per K) differs from the induced

orientation they inherit from C_ℓ . Notice that because the length of C_ℓ is even, the parity of $n(C_\ell)$ is independent of how we orient C_ℓ (this is why our choice of clockwise didn't really matter). Putting everything together, we have

$$\varepsilon(D)\varepsilon(D') = (-1)^m \prod_{\ell=1}^m (-1)^{n(C_\ell)} = \prod_{\ell=1}^m (-1)^{n(C_\ell)+1}.$$

and we see that it shall indeed be the case that $\varepsilon(D)\varepsilon(D') = 1$ if $n(C_\ell)$ is odd for each ℓ .

Notice, however, that this condition isn't super easy to enforce in practice; namely, in the sense that it requires one to examine each symmetric difference $D\Delta D'$ separately to find the curves C_ℓ . It would be easier if we had a way of finding these curves right from the get-go. This is exactly what the following lemma does for us.

Lemma 4.2.3. *A cycle C of even length in G arises in the symmetric difference $D\Delta D'$ for some $D, D' \in \mathcal{D}$ if and only if $G \setminus C$ admits a dimer configuration.*

Proof. It is clear that if C arises in some symmetric difference $D\Delta D'$, then $G \setminus C$ admits a dimer configuration. Conversely, suppose that C is such that $G \setminus C$ admits a dimer configuration D . Pick an edge e_1 in C and, moving clockwise, enumerate the next edge e'_1 , the one after that e_2 , then e'_2 and so forth, so that $C = \{e_1, e'_1, \dots, e_k, e'_k\}$ and e'_ℓ is adjacent to e_ℓ for each ℓ . Then $D' := D \cup \{e_1, \dots, e_k\}$ and $D'' := D \cup \{e'_1, \dots, e'_k\}$ are two dimer configurations of G for which $C \in D'\Delta D''$. $\square$

The above lemma tells us that to prove [Theorem 4.2.1](#), it is sufficient to choose K so that for any cycle C of even length in G such that $G \setminus C$ admits a dimer configuration, $n(C)$ is odd.

Lemma 4.2.4. *This is possible.*

Proof. Let us say that an orientation K satisfies the *Kasteleyn condition* around a face f of G if $n(\partial f)$ is odd. An orientation which satisfies the Kasteleyn condition around each face shall be called a *Kasteleyn orientation*. The proof proceeds in two steps.

Step 1: Any Kasteleyn orientation K of G has the desired property that for any cycle C of even length in G such that $G \setminus C$ admits a dimer configuration, $n(C)$ is odd. Let C be any such cycle. Then C must enclose an even number of vertices V_{int} (as these are matched into pairs by D) and some (even or odd) number of edges E_{int} (here, we also include edges that join internal vertices to vertices which lie on C). Note that by assumption, C itself consists of an even number of edges E_C and an equal number of vertices V_C . Let $E = E_{int} + E_C$ and $V = V_{int} + V_C$, so that the subgraph S determined by C and its interior consists of E edges and V vertices. Moreover, let $\mathcal{F}$ denote set of faces enclosed by C and F its cardinality. Then since $n(\partial f)$ is odd for every $f \in \mathcal{F}$ (as K is a Kasteleyn orientation), we have the modulo 2 equality

$$0 \equiv \sum_{f \in \mathcal{F}} (n(\partial f) + 1) = \sum_{f \in \mathcal{F}} n(\partial f) + F = n(C) + E_{int} + F.$$

Above, the E_{int} arises in the third equality because each internal edge in E_{int} lies on the boundary of two faces in $\mathcal{F}$; since traversing one of them clockwise amounts to traversing the boundary of the other counter-clockwise, this edge contributes $+1$ to the sum. From this, together with the fact that $E_C = V_C$ and V_{int} are even, we furthermore get (again, modulo 2)

$$0 \equiv n(C) + (E_C + E_{int}) + (V_C + V_{int}) + F = n(C) + E + V + F \equiv n(C) - E + V + F.$$

Since $F - E + V = 1$ by Euler's formula, it follows from the above that $n(C) \equiv 1$ modulo 2 and hence that $n(C)$ is odd.

Step 2: There exists a Kasteleyn orientation K of G . Fix a spanning tree $T \in G^*$, rooted at the vertex corresponding to the outer face of G . Orient arbitrarily the edges of G which do not intersect T . Now move along T , beginning from the leaves and proceeding towards the root, and remove the edges one by one and orient the corresponding edge of G as follows: if the number of clockwise co-oriented edges along the boundary of the current face f is odd, then orient the edge so that it agrees with a clockwise orientation of ∂f ; otherwise, orient it oppositely. Once all the edges of T are removed, it is easy to see that we are left with an orientation K which satisfies the Kasteleyn condition around each face of G . $\square$

Corollary 4.2.1. *Theorem 4.2.1 is true.* $\square$

If, in addition, our graph G is bipartite (2-colorable), then the dimer partition function $Z(G; \nu)$ can be expressed in an even more simplistic fashion. Indeed, if we color the $2n$ vertices of G black and white, so that there are n black vertices and n white vertices, and then label the vertices $1, \dots, 2n$ so that the first vertices $1, \dots, n$ are white and vertices $n+1, \dots, 2n$ are black, then any corresponding weighted skew-adjacency matrix A is of the form

$$A = \begin{pmatrix} 0 & M \\ -M^\top & 0 \end{pmatrix}$$

where M is $n \times n$. Now, since

$$A = \begin{pmatrix} M & 0 \\ 0 & -M^\top \end{pmatrix} \cdot \begin{pmatrix} 0 & I \\ I & 0 \end{pmatrix},$$

and the determinant of the latter is easily seen to be $(-1)^n$, it follows via [Corollary 4.1.1](#) that

$$\text{pf}(A)^2 = \det(A) = (-1)^n \det(M) \det(-M^\top) = \det(M)^2.$$

Therefore, by [Theorem 4.2.1](#), we can choose K so that $Z(G; \nu) = |\det M|$. Unfortunately, there is little hope to extend [Theorem 4.2.1](#) to non-planar graphs since many such graphs do not admit a Pfaffian orientation, the simplest example being $K_{3,3}$.

Chapter 5

The Ising Model

In this chapter, we shall discuss several special instances of a wider class of models known as *Ising models*, which can be thought of as models of magnets. Ising models regard the molecules which make up a magnet as points on a rectangular lattice. For illustrative purposes, suppose that there are N such molecules (and hence sites on the lattice), labelled $i = 1, \dots, N$. We regard each molecule as a microscopic magnet which points either along some preferred axis or in exactly the opposite direction. Each molecule i thus has two possible configurations, which we opt to label by a “spin” variable $\sigma_i \in \{\pm 1\}$; $+1$ if the molecule points parallel to the preferred axis, and -1 if it points anti-parallel to the preferred axis. The spin is said to be “up” when σ_i has value $+1$ and “down” when it has value -1 , and we often abbreviate these values to simply $+$ and $-$, respectively.

If we let $\sigma := \{\sigma_1, \dots, \sigma_N\}$ denote the set of all N spins, then there are 2^N values of σ , each of which specifies a state of the system (magnet). The Hamiltonian E of the system is then a function of σ , and can be decomposed into two parts:

$$E(\sigma) = E_0(\sigma) + E_1(\sigma), \quad (5.1)$$

where E_0 denotes the contribution from the intermolecular forces inside the magnet and E_1 the contribution from the interactions between the molecules and an external magnetic field. The latter can be written as

$$E_1(\sigma) = -H \sum_{i=1}^N \sigma_i, \quad (5.2)$$

where H is proportional to the component of the magnetic field in the direction of the preferred axis.

Because we also expect interactions within a physical system to be “invariant under time reversal,” meaning that E should be unchanged if we negate each of H and $\sigma_1, \dots, \sigma_N$, it is an immediate consequence of the relation [Eq. \(5.2\)](#) and the equation defining E that E_0 must be an even function of σ ; that is,

$$E_0(\sigma_1, \dots, \sigma_N) = E_0(-\sigma_1, \dots, -\sigma_N). \quad (5.3)$$

The two relations [Eq. \(5.2\)](#) and [Eq. \(5.3\)](#) define the class of models known as Ising models, several special instances of which have been exactly solved. Just as with the Dimer model, we can associate to each Ising model a *partition function* Z , given by

$$Z(N, H, T) = \sum_s \exp[-E(s)/kT] = \sum_{\sigma} \exp[-(E_0(\sigma) - H \sum_{i=1}^N \sigma_i)/kT] \quad (5.4)$$

where the summation extends over all states s of the system, k is an unimportant constant related to the physics of such systems, and (similar to H) T is a physical parameter that one can interpret as the temperature of the system.

Also of importance to Ising models is the so-called “free energy” of the system, which is defined to be the following “thermodynamic limit”:

$$f(H, T) = -kT \lim_{N \rightarrow \infty} \frac{\log Z(N, H, T)}{N}.$$

For unimportant physical reasons, we expect this limit to always exist. We also define the “magnetization” M of the system as

$$M(H, T) = -\frac{\partial}{\partial H} f(H, T).$$

The goal of the following sections will be to compute Z , f , and M exactly in several special cases.

5.1 The One-Dimensional Ising Model

Consider an Ising model on a line of N sites, labelled successively $i = 1, \dots, N$, with internal energy E_0 given by nearest-neighbour interaction:

$$E_0(\sigma) = -J \sum_{j=1}^N \sigma_j \sigma_{j+1}.$$

Here, J is a nonzero constant and site N is regarded as being followed by site 1, so that $\sigma_{N+1} = \sigma_1$. By Eq. (5.4), the partition function for the model is then given by

$$Z = \sum_{\sigma} \exp \left[K \sum_{j=1}^N \sigma_j \sigma_{j+1} + h \sum_{j=1}^N \sigma_j \right]$$

where $K = J/kT$ and $h = H/kT$. Note that we can simply the above into summands involving only two neighbouring spins by factoring the exponential term:

$$Z = \sum_{\sigma} \exp \left[K \sum_{j=1}^N \sigma_j \sigma_{j+1} + \frac{1}{2} h \sum_{j=1}^N (\sigma_j + \sigma_{j+1}) \right] = \sum_{\sigma} \prod_{j=1}^N \exp \left[K \sigma_j \sigma_{j+1} + \frac{1}{2} h (\sigma_j + \sigma_{j+1}) \right].$$

This is can equivalently be expressed as

$$Z = \sum_{\sigma} \prod_{j=1}^N V(\sigma_j, \sigma_{j+1}) = \sum_{\sigma_1} \cdots \sum_{\sigma_N} V(\sigma_1, \sigma_2) \cdots V(\sigma_{N-1}, \sigma_N) V(\sigma_N, \sigma_1)$$

where $V(\sigma, \sigma') = \exp \left[K \sigma \sigma' + \frac{1}{2} h (\sigma + \sigma') \right]$. The key observation leading to solvability of Z is that if we then encode the possible values of $V(\sigma, \sigma')$ as the entries of a 2×2 matrix

$$V = \begin{pmatrix} V(+, +) & V(+, -) \\ V(-, +) & V(-, -) \end{pmatrix} = \begin{pmatrix} e^{K+h} & e^{-K} \\ e^{-K} & e^{K-h} \end{pmatrix},$$

then the summations over $\sigma_2, \dots, \sigma_N$ in the above expression for Z can be regarded as successive matrix multiplications, and the summation over σ_1 the taking of a trace, so that

$$Z = \text{tr}(V^N).$$

Finally, if we let $a = e^{K+h}$, $b = e^{K-h}$, and $c = e^{-K}$, then because

$$\chi_V(\lambda) = \det(\lambda I - V) = \lambda^2 - (a+b)\lambda + (ab - c^2)$$

has solution

$$\begin{aligned} \lambda &= \frac{1}{2} \left[(a+b) \pm \sqrt{(a+b)^2 - 4(ab - c^2)} \right] = e^K \cosh(h) \pm \frac{1}{2} \sqrt{4e^{2K} \cosh^2(h) - 4(e^{2K} - e^{-2K})} \\ &= e^K \cosh(h) \pm \sqrt{e^{2K} [\cosh^2(h) - 1] + e^{-2K}} \\ &= e^K \cosh(h) \pm \sqrt{e^{2K} \sinh^2(h) + e^{-2K}}, \end{aligned}$$

it follows that

$$Z = \left(e^K \cosh(h) - \sqrt{e^{2K} \sinh^2(h) + e^{-2K}} \right)^N + \left(e^K \cosh(h) + \sqrt{e^{2K} \sinh^2(h) + e^{-2K}} \right)^N.$$

Next, we note that because the roots λ_+, λ_- of χ_V are the sum and difference (respectively) of two positive numbers, we have that $\lambda_+ > \lambda_-$ and consequently that $|\lambda_-/\lambda_+| < 1$. Since

$$Z = \lambda_-^N + \lambda_+^N = \lambda_+^N [(\lambda_-/\lambda_+)^N + 1],$$

it then follows that

$$\frac{\log Z}{N} = \log \lambda_+ + \frac{\log[(\lambda_-/\lambda_+)^N + 1]}{N} \rightarrow \log \lambda_+ \text{ as } N \rightarrow \infty.$$

We therefore have that

$$f(H, T) = -kT \log \left(e^K \cosh(h) + \sqrt{e^{2K} \sinh^2(h) + e^{-2K}} \right)$$

and hence (after a little simplification) that

$$M(H, T) = \frac{e^K \sinh(h)}{\sqrt{e^{2K} \sinh^2(h) + e^{-2K}}}.$$

5.2 The Square-Lattice Ising Model

Consider an Ising model a square lattice $\mathcal{L}$ with m rows and n columns. To each site $i = (i_1, i_2)$, there is an associated spin variable $\sigma_i \in \{\pm 1\}$; let us define E_0 by saying that two nearest-neighbour spins σ_i, σ_j contribute $-J\sigma_i\sigma_j$ to the Hamiltonian E if they are horizontal neighbours and $-J'\sigma_i\sigma_j$ if they are vertical neighbours. If, in addition, we stipulate that $E_1 \equiv 0$ (i.e., there is no external magnetic field), then it follows that the partition function $Z(N, H, T)$ (which, for the sake of brevity, shall henceforth be denoted by Z_N) is given by

$$Z_N = \sum_{\sigma} \exp \left[K \sum_{(i,j)} \sigma_i \sigma_j + L \sum_{(i,k)} \sigma_i \sigma_k \right],$$

where $K = J/kT$, $L = J'/kT$, and the first summation extends over all horizontal edges in $\mathcal{L}$ while the second extends over all vertical edges in $\mathcal{L}$. The goal of this section will be to exactly compute the free energy

$$f = -kT \lim_{N \rightarrow \infty} \frac{\log Z_N}{N}.$$

5.2.1 Transfer Matrices and the Partition Function

One can imagine rotating the lattice $\mathcal{L}$ so that it is now oriented diagonally, as in Fig. 5.1. Then our expression for Z_N is still valid, provided that the first summation now extend over all edges parallel to those marked K in Fig. 5.1, and the second summation all edges parallel to those marked L .

Let us now group the sites into horizontal rows based on two types A and B : “solid circles” and “open circles”, respectively, again as depicted in Fig. 5.1. Let us perform this grouping so that a row of type A is always above a row of type B , and vice-versa; that is, the rows should alternate types. Now label the rows $1, \dots, m$ so that row r is below row $r+1$, and impose cyclic boundary conditions so that row m is thought of as being below row 1. Note that because we require that the rows alternate between the types A and B , this means that m must be even. Similarly, for each individual row, let us label the sites $1, \dots, n$ from left to right, again imposing cyclic boundary conditions so that site n is to the left of site 1.

Let ϕ_r denote the set of all spins in row r , so that ϕ_r has 2^n possible values. If $\phi = \{\sigma_1, \dots, \sigma_n\}$ and $\phi' = \{\sigma'_1, \dots, \sigma'_n\}$ are let to denote two successive rows (lower and upper, respectively), and we define

$$V_{\phi, \phi'} = \exp \left[\sum_{j=1}^n (K \sigma_{j+1} \sigma'_j + L \sigma_j \sigma'_j) \right]$$

$$W_{\phi, \phi'} = \exp \left[\sum_{j=1}^n (K \sigma_j \sigma'_j + L \sigma_j \sigma'_{j+1}) \right],$$

then it is not hard to see that Z_N factorizes as

$$Z_N = \sum_{\phi_1} \cdots \sum_{\phi_m} V_{\phi_1, \phi_2} W_{\phi_2, \phi_3} \cdots V_{\phi_{m-1}, \phi_m} W_{\phi_m, \phi_1}.$$

As with the single dimensional case, if we encode the values $V_{\phi, \phi'}, W_{\phi, \phi'}$ as the entries of $2^n \times 2^n$ matrices V and W , respectively, then the summations over $\phi_2, \dots, \phi_m$ can be regarded as successive matrix multiplications, and the summation over ϕ_1 as the taking of a trace, so that the above reduces to

$$Z_N = \text{tr} VWVW \cdots VW = \text{tr}(VW)^{m/2}.$$

Equivalently, we have $Z_N = \Lambda_1^m + \dots + \Lambda_{2^n}^m$, where the Λ_j^2 denote the eigenvalues of VW repeated according to multiplicity. It follows at once that $Z_N \sim \Lambda_{\max}^m$, where $\Lambda_{\max}$ denotes the eigenvalue of maximum modulus. Since we are interested in the limit of $\log Z_N/N$ as $N \rightarrow \infty$, the goal is now to compute $\Lambda_{\max}$.

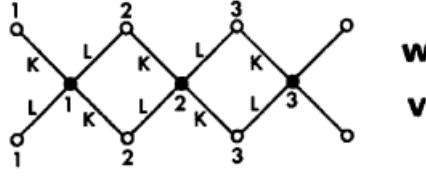


Figure 5.1: Three successive rows of the lattice $\mathcal{L}$

How are we going to do this? From the equations defining $V_{\phi, \phi'}$ and $W_{\phi, \phi'}$, it is clear that V and W can be thought of as functions of K and L (the “interaction coefficients”). The strategy will be to establish some nice properties of these matrices, as functions of K and L (which, for our purposes, shall be thought of as complex parameters), that enable $\Lambda_{\max}$ to be computed.

5.2.2 The Star-Triangle Relation

As per our calculation from the previous subsection, one is (in computing Z_N) interested in the product $VW = V(K, L)W(K, L)$. Let us generalize our perspective and instead consider the product $V(K, L)W(K', L')$. This product is the transfer matrix for the lattice with rows depicted in Fig. 5.1, but with the interaction coefficients K, L of the edges above the solid circles replaced by K' and L' . If we let $\phi = \{\sigma_1, \dots, \sigma_n\}$ denote the set of spins on the lower row of open circles, and $\phi' = \{\sigma'_1, \dots, \sigma'_n\}$ the set of spins on the upper row, then it follows that the (ϕ, ϕ') entry of the product $V(K, L)W(K', L')$ is given by

$$\sum_{\sigma'_1} \dots \sum_{\sigma'_n} \prod_{j=1}^n \exp [\sigma''_j (K\sigma_{j+1} + L\sigma_j + K'\sigma'_j + L'\sigma'_{j+1})],$$

where the summations extend over all spins $\sigma''_1, \dots, \sigma''_n$ of the intermediate row of solid circles. Above, the exponential term is the product of the so-called “Boltzmann weights” of the four edges with solid circle j as an endpoint. Since σ''_j enters only this term, we can factor the above as

$$\prod_{j=1}^n X(\sigma_j, \sigma_{j+1}; \sigma'_j, \sigma'_{j+1}),$$

where, for $a, b, c, d \in \{\pm 1\}$, $X(a, b; c, d) = \sum_{f=\pm 1} \exp[f(La + Kb + L'c + K'd)]$.

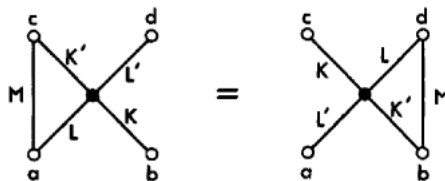
We now ask: if we interchange K with K' and L with L' , does this change the product matrix $V(K, L)W(K', L')$? In other words, we ask whether the following “commutativity” relation holds:

$$V(K, L)W(K', L') = V(K', L')W(K, L). \quad (5.5)$$

Clearly, the value of $\prod_{j=1}^n X(\sigma_j, \sigma_{j+1}; \sigma'_j, \sigma'_{j+1})$ is not changed if we replace each factor $X(a, b; c, d)$ with the new factor $e^{Mac}X(a, b; c, d)e^{-Mbd}$ for a fixed number M , since our cyclic boundary conditions ensure that the exponential factors from adjacent terms will cancel. It follows, then, that Eq. (5.5) shall hold if there is a number M such that

$$e^{Mac}X(a, b; c, d) = X'(a, b; c, d)e^{Mbd},$$

where $X'(a, b; c, d)$ is obtained from X by interchanging K with K' and L with L' . Geometrically, this means that the Boltzmann weights of the following diagrams must be equal for all a, b, c, d :



To simplify this condition to a condition involving only the parameters K, L, K', L' , we can use the so-called “star-triangle relation”, which is explicit relationship between the partition functions Z_N^T, Z_N^H for triangular and hexagonal lattices, respectively. It proceeds as follows. We first note that the partition function for a hexagonal lattice is given by

$$Z_N^H = \sum_{\sigma} \exp \left[L_1 \sum_{(i,\ell)} \sigma_i \sigma_{\ell} + L_2 \sum_{(j,\ell)} \sigma_j \sigma_{\ell} + L_3 \sum_{(k,\ell)} \sigma_k \sigma_{\ell} \right],$$

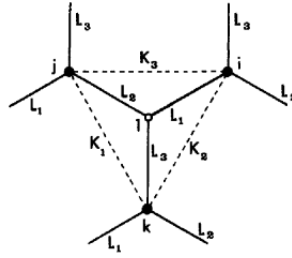
where L_1, L_2, L_3 are nearest-neighbour energy constants associated with the sets of edges parallel to each set of opposite edges of a regular hexagon. Similarly, for a triangular lattice we have

$$Z_N^T = \sum_{\sigma} \exp \left[K_1 \sum_{(j,k)} \sigma_j \sigma_k + K_2 \sum_{(k,i)} \sigma_k \sigma_i + K_3 \sum_{(i,j)} \sigma_i \sigma_j \right].$$

Now, because a hexagonal lattice is clearly bipartite, we can assign one of the types A and B we defined earlier to each site in the lattice so that each site of type A is adjacent only to sites of type B and vice-versa. If we do this, then we may write

$$Z_N^H = \sum_{\sigma} \prod_{\ell} W(\sigma_{\ell} | \sigma_i, \sigma_j, \sigma_k),$$

where $W(\sigma_{\ell} | \sigma_i, \sigma_j, \sigma_k) = \exp[\sigma_{\ell}(L_1 \sigma_i + L_2 \sigma_j + L_3 \sigma_k)]$ and the product extends over all B -sites ℓ , with A -site nearest-neighbours i, j , and k , as per the following diagram.



Because each B -spin σ_{ℓ} occurs in one and only one factor, we can interchange the product and summation over the B -sites, yielding

$$Z_N^H = \sum_{\sigma_A} \prod_{(i,j,k)} w(\sigma_i, \sigma_j, \sigma_k),$$

where the summation now extends over all A -sites σ_A , and

$$w(\sigma_i, \sigma_j, \sigma_k) = \sum_{\sigma_{\ell}} W(\sigma_{\ell} | \sigma_i, \sigma_j, \sigma_k) = 2 \cosh(L_1 \sigma_i + L_2 \sigma_j + L_3 \sigma_k).$$

Finally, we note that because $w(\sigma_i, \sigma_j, \sigma_k)$ is unaltered if we negate each of $\sigma_i, \sigma_j, \sigma_k$ (since $\cosh$ is an even function), and since $\sigma_i, \sigma_j, \sigma_k \in \{\pm 1\}$, there exist parameters R, K_1, K_2, K_3 such that

$$w(\sigma_i, \sigma_j, \sigma_k) = R \exp[K_1 \sigma_j \sigma_k + K_2 \sigma_k \sigma_i + K_3 \sigma_i \sigma_j],$$

from which it follows that (since the collection of A -sites form a triangular lattice)

$$Z_N^H = R^{N/2} \sum_{\sigma_A} \prod_{(i,j,k)} \exp[K_1 \sigma_j \sigma_k + K_2 \sigma_k \sigma_i + K_3 \sigma_i \sigma_j] = R^{N/2} Z_{N/2}^T \iff Z_{2N}^H = R^N Z_N^T.$$

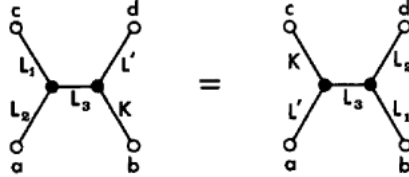
This is the so-called “star-triangle” relation.

Recall, now, the geometric condition we derived earlier to ensure that [Eq. \(5.5\)](#) holds. This condition was that the Boltzmann weights of two diagrams be equal. Note, however, that in each of the diagrams there is a

(K', L, M) triangle. If we let $K_1 = L$, $K_2 = K'$, and $K_3 = M$, then by the star-triangle relation there exist parameters R, L_1, L_2, L_3 , defined by

$$\begin{aligned} 2 \cosh(L_1 + L_2 + L_3) &= R \exp(K_1 + K_2 + K_3) \\ 2 \cosh(-L_1 + L_2 + L_3) &= R \exp(K_1 - K_2 - K_3) \\ 2 \cosh(L_1 - L_2 + L_3) &= R \exp(-K_1 + K_2 - K_3) \\ 2 \cosh(L_1 + L_2 - L_3) &= R \exp(-K_1 - K_2 + K_3), \end{aligned}$$

so that the Boltzmann weights of the previous figures are (up to the factor R), equal to the Boltzmann weights of the following figures:



It follows, then, that for Eq. (5.5) to hold, it shall suffice for the Boltzmann weights of the above diagrams be equal. Clearly this shall be the case if $L_1 = K$ and $L_2 = L'$. If we make these substitutions into the four equations above (defining the parameters R, L_2, L_2, L_3) and eliminate R, L_3 , and M , one finds that this shall be true provided that

$$\sinh(2K) \sinh(2L) = \sinh(2K') \sinh(2L').$$

We have thus derived a purely algebraic condition on the parameters K, L, K', L' to ensure that Eq. (5.5) holds.

5.2.3 Inversion

Given K and L , we now ask whether K' and L' can be chosen so that the product $V(K, L)W(K', L')$ is diagonal or “near-diagonal.” Recall that this product matrix has (ϕ, ϕ') -entry

$$\prod_{j=1}^n X(\sigma_j, \sigma_{j+1}; \sigma'_j, \sigma'_{j+1}),$$

where the σ_j denote the spins in row ϕ and the σ'_j denote the spins in row ϕ' . It is not hard to see, then, that $V(K, L)W(K', L')$ would be diagonal if $X(a, b; c, d) = 0$ whenever $a \neq c$ or $b \neq d$. It turns out, however, that this condition is too strong. What can be ensured, however, is that $X(a, b; c, d)$ vanish if $a \neq c$ and $b = d$ (i.e., we can make the matrix “near-diagonal”). Indeed, since $X(a, b; c, d) = \sum_{f=\pm 1} \exp[f(La + Kb + K'c + L'd)] = 2 \cosh(La + Kb + K'c + L'd)$, this is equivalent to requiring that

$$\cosh(L + K - K' + L') = 0, \quad \cosh(L - K - K' - L') = 0.$$

It is straightforward to verify that this system of equations has complex solution $K' = L + \frac{1}{2}i\pi$ and $L' = -K$.

Choosing K' and L' in this way (so that $X(a, b; c, d) = 0$ if $a \neq c$ and $b = d$) means that for nonzero elements of $V(K, L)W(K', L')$, if σ_j and σ'_j are unlike, then σ_{j+1} and σ'_{j+1} must also be unlike. Because of our cyclic boundary conditions, it follows that either all pairs (σ_j, σ'_j) are like, or they are all unlike. If the former is true, then in every factor $X(a, b; c, d)$ we have $a = c$ and $b = d$. From the definition of X and our choice of K' and L' , one finds that the only possible such factor is $2i \sinh 2L$; thus, the corresponding entry of the matrix is $(2i \sinh 2L)^n$. Similarly, if the latter is true then every factor $X = -2i \sinh 2K$ and the corresponding matrix entry is $(-2i \sinh 2K)^n$. In general, it follows that the (ϕ, ϕ') entry of $V(K, L)W(K', L')$ is

$$\prod_{j=1}^n X(\sigma_j, \sigma_{j+1}; \sigma'_j, \sigma'_{j+1}) = (2i \sinh 2L)^n \delta(\sigma_1, \sigma'_1) \cdots \delta(\sigma_n, \sigma'_n) + (-2i \sinh 2K)^n \delta(\sigma_1, -\sigma'_1) \cdots \delta(\sigma_n, -\sigma'_n).$$

Equivalently, if I is let to denote the $2^n \times 2^n$ identity matrix and R the $2^n \times 2^n$ matrix with (ϕ, ϕ') -entry

$$R_{\phi, \phi'} = \delta(\sigma_1, -\sigma'_1) \cdots \delta(\sigma_n, -\sigma'_n),$$

then we have that

$$V(K, L)W(L + \tfrac{1}{2}i\pi, -K) = (2i \sinh 2L)^n I + (-2i \sinh 2K)^n R. \quad (5.6)$$

Because $R^2 = I$, the right-hand side above is easily inverted (multiplying by the conjugate gives a scalar multiple of I), and so one could easily use this identity to find an inverse for the matrix $V(K, L)$. For this reason, we call it the “inversion identity.”

5.2.4 Symmetry

Recall that original matrices $V = V(K, L)$ and $W = W(K, L)$ have entries

$$V_{\phi, \phi'} = \exp \left[\sum_{j=1}^n (K\sigma_{j+1}\sigma'_j + L\sigma_j\sigma'_j) \right]$$

$$W_{\phi, \phi'} = \exp \left[\sum_{j=1}^n (K\sigma_j\sigma'_j + L\sigma_j\sigma'_{j+1}) \right].$$

It is easy to see from this that negating K and L has the same effect (for both V and W) as negating each of $\sigma_1, \dots, \sigma_n$ or each of $\sigma'_1, \dots, \sigma'_n$. Since the former is achieved by multiplication on the left by R and the latter multiplication on the right by R , we equivalently have that

$$V(-K, -L) = RV(K, L) = V(K, L)R \quad (5.7)$$

and similarly for W . This is our first “symmetry” identity. Given a fixed pair of rows ϕ, ϕ' , let r denote the number of unlike pairs of spins $(\sigma_{j+1}, \sigma'_j)$ and s the number of unlike pairs (σ_j, σ'_j) . Clearly, $r + s$ counts the number of changes of sign in the sequence

$$\sigma_1, \sigma'_1, \dots, \sigma_n, \sigma'_n$$

so by cyclic boundary conditions $r + s$ must be even number. Moreover, from the above equation for $V_{\phi, \phi'}$ we have

$$V_{\phi, \phi'} = \exp[(n - 2r)K + (n - 2s)L]$$

since each unlike pair $(\sigma_{j+1}, \sigma'_j)$ contributes two -1 terms to the sum, and similarly for the unlike pairs (σ_j, σ'_j) . We are interested in the case of n large; since it does not matter how n gets large, let us assume that $n = 2p$ for some positive integer p . Then the above reduces to

$$V_{\phi, \phi'} = \exp[\pm 2r'K \pm 2s'L], \quad (5.8)$$

where $r' := p - r$ and $s' := p - s$ are integers in $(0, p)$. Now, because $r + s$ is even, r and s must have the same parity. It follows that r' and s' must also be of the same parity and consequently that the value of

$$V_{\phi, \phi'} = \exp[\pm 2r'K \pm 2s'L] = \exp(2K)^{\pm r'} \exp(2L)^{\pm s'}$$

is unchanged if we negate each of $\exp(2K)$ and $\exp(2L)$. Since this achieved by replacing K with $K + \frac{1}{2}\pi i$ and L with $L + \frac{1}{2}\pi i$, we derive another “symmetry” identity

$$V(K, L) = V(K + \tfrac{1}{2}i\pi, L + \tfrac{1}{2}i\pi). \quad (5.9)$$

5.2.5 Commutativity

Recall that we showed [Eq. \(5.5\)](#) to be true provided that $\sinh(2K) \sinh(2L) = \sinh(2K') \sinh(2L')$. We now show that this condition in fact implies that the matrices

$$V(K, L), W(K, L), V(K', L'), W(K', L')$$

all commute. Define C to be the $2^n \times 2^n$ matrix with (ϕ, ϕ') entry given by

$$\delta(\sigma_1, \sigma'_2) \delta(\sigma_2, \sigma'_3) \cdots \delta(\sigma_n, \sigma'_1).$$

Then the transformation $A \mapsto C^{-1}AC$ has the effect of changing the spin labels from $1, \dots, n$ to $2, \dots, n, 1$. Because of our cyclic boundary conditions, it leaves V and W unchanged. In other words, for all parameters K, L, K', L' we have

$$V = C^{-1}VC, \quad W = C^{-1}WC$$

meaning that V and W each commute with C . It is furthermore straightforward to check that $W(K, L) = V(K, L)C$ (and similarly for K', L'), so in fact $V(K, L)$ (resp. $V(K', L')$) and $W(K, L)$ (resp. $W(K', L')$) commute.

If we now sub $W = VC$ into the relation $V(K, L)W(K', L') = V(K', L')W(K, L)$ (recall that we are assuming it to be the case that $\sinh(2K)\sinh(2L) = \sinh(2K')\sinh(2L')$) and use the fact that V commutes with C , we derive easily that

$$V(K, L)V(K', L') = V(K', L')V(K, L);$$

that is, $V(K, L)$ and $V(K', L')$ commute. Since $W = VC$, it immediately follows from this that $W(K, L)$ and $W(K', L')$ likewise commute. In fact, because $V(K, L)$ commutes with $V(K', L')$ and C , and $W(K', L') = V(K', L')C$, $V(K, L)$ also commutes with $W(K', L')$ and similarly $V(K', L')$ commutes with $W(K, L)$. In other words, the matrices $V(K, L), W(K, L), V(K', L'), W(K', L')$ all commute, as claimed.

We now note that because $W = VC$, we can eliminate W from the inversion identity Eq. (5.6) to get

$$V(K, L)V(L + \tfrac{1}{2}i\pi, -K)C = (2i \sinh 2L)^n I + (-2i \sinh 2K)^n R. \quad (5.10)$$

Finally, we note that because the transformation $A \mapsto R^{-1}AR$ has the effect of negating all spins

$$\sigma_1, \dots, \sigma_n, \sigma'_1, \dots, \sigma'_n,$$

and this clearly leaves V and W unchanged, each of V and W commute with R .

5.2.6 Eigenvalues

Let $k = (\sinh 2K \sinh 2L)^{-1}$ and regard K and L as complex numbers subject to this constraint. By varying K and L , we generate an infinite set of transfer matrices V which by the previous subsection all commute and thus share a common set of eigenvectors. Let x denote one such eigenvector, and let $v(K, L)$, c , and r denote the corresponding eigenvalues of $V(K, L)$, C , and R , respectively. Note that because $C^n = R^2 = I$, we have $c^n = r^2 = 1$.

Applying both sides of the inversion identity Eq. (5.10) to x and equating coefficients, one finds that

$$v(K, L)v(L + \tfrac{1}{2}i\pi, -K)c = (2i \sinh 2L)^n + (-2i \sinh 2K)^n r.$$

Recall that we showed $Z_N = \sum_j \Lambda_j^m$, where the λ_j^2 are the eigenvalues of VW , repeated according to multiplicity. Since x must also be an eigenvector of VW , let $\Lambda(K, L)$ denote the corresponding Λ_j , so that (since $W = VC$)

$$\Lambda^2(K, L) = v^2(K, L)c \iff \Lambda(K, L) = v(K, L)c^{1/2}.$$

Using this, the previous identity can be written in the c -independent form

$$\Lambda(K, L)\Lambda(L + \tfrac{1}{2}i\pi, -K) = (2i \sinh 2L)^n + (-2i \sinh 2K)^n r. \quad (5.11)$$

Together with some simple analytic properties of $\Lambda(K, L)$ as a function of the complex variables K and L , we now show that this relation determines $\Lambda(K, L)$ completely.

We address only the case $k = 1$ (the case of general k can be done following a similar programme, however is much more contrived). Then, rather than K and L , it shall be convenient to work with a variable u defined by the following system of equations:

$$\begin{aligned} \sinh 2K &= \tan u \\ \sinh 2L &= \cot u. \end{aligned}$$

Then $\Lambda(K, L)$ can be thought of as a function of u , and our previous identity Eq. (5.11) becomes

$$\Lambda(u)\Lambda(u + \tfrac{1}{2}\pi) = (2i \cot u)^n + (-2i \tan u)^n r.$$

The key benefit of working with a single variable is that $\exp(\pm 2K)$ and $\exp(\pm 2L)$ can be expressed as simple functions of u . For example, because

$$\sinh 2K = \tan u \iff 1 - \cosh^2(2K) = \tan^2(u) \iff \cosh^2(2K) = 1 - \tan^2(u) = \sec^2(u)$$

we have

$$\exp(2K) + \exp(-2K) = 2 \cosh 2K = 2 \sec u, \quad \exp(2K) - \exp(-2K) = 2 \sinh 2K = 2 \tan u.$$

Solving this system of equations for each of $\exp(2K)$ and $\exp(-2K)$, one finds that

$$\exp(2K) = \frac{1 + \sin u}{\cos u}, \quad \exp(-2K) = \frac{1 - \sin u}{\cos u}.$$

Similarly, we have

$$\exp(2L) = \frac{1 + \cos u}{\sin u}, \quad \exp(-2L) = \frac{1 - \cos u}{\sin u}.$$

The important thing to note is that each of these functions is single-valued, meromorphic, and periodic with period 2π . Subbing these identities into Eq. (5.8), we find that

$$\begin{aligned} V_{\phi, \phi'} &= [\exp(\pm 2K)]^{r'} [\exp(\pm 2L)]^{s'} = \left(\frac{1 \pm \sin u}{\cos u} \right)^{r'} \left(\frac{1 \pm \cos u}{\sin u} \right)^{s'} \\ &= (\sin u \cos u)^{-p} \cdot (1 \pm \sin u)^{r'} (\cos u)^{p-r'} (1 \pm \cos u)^{s'} (\sin u)^{p-s'}. \end{aligned}$$

In other words, we have that

$$V_{\phi, \phi'} = \frac{t(u)}{(\sin u \cos u)^p},$$

where $t(u)$ is a polynomial in $\sin u$ and $\cos u$ of degree $2p$. Any such polynomial can be written in the form

$$t(u) = e^{-2ipu} (c_0 + c_1 e^{iu} + \dots + c_{2p} e^{4ipu}). \quad (5.12)$$

Let us now return our attention back to the vector equation $V(K, L)x = v(K, L)x$. Really, it is a collection of 2^n scalar equations, each one of which expresses the eigenvalue $v(K, L)$ as a linear combination of the entries of $V(K, L)$, the coefficients being ratios of the elements of x . Because of the commutativity relations we derived however, these ratios cannot depend on u (as varying u is equivalent to varying K and L subject to the constraint $k = 1$, but neither of these changes shall effect x). It follows that $v(K, L)$ is a linear combination of functions of the form $t(u)/(\sin u \cos u)^p$ with constant coefficients, and hence is itself of this form. If we now recall the identity $\Lambda(K, L) = v(K, L)c^{1/2}$ we derived earlier, we see that in fact $\Lambda(K, L) = \Lambda(u)$ must also be of this form.

Using the symmetry relations we derived earlier, this form can be simplified further. For example, if we replace u with $u + \pi$, then by the above identities for $\exp(\pm 2K)$ and $\exp(\pm 2L)$, this has the same effect as replacing K and L by $-K \pm \frac{1}{2}i\pi$ and $-L \pm \frac{1}{2}i\pi$. By the symmetry identities Eq. (5.7) and Eq. (5.9), this in turn is the same as multiplying V by R . In other words, if we write $v(K, L)$ as $v(u)$, then the equation $V(K, L)x = v(K, L)x$ becomes

$$V(K, L)Rx = v(u + \pi)x \implies rv(u)x = v(u + \pi)x \implies v(u + \pi) = rv(u) \implies \Lambda(u + \pi) = r\Lambda(u),$$

where the last implication follows from the fact that $\Lambda(u) = v(u)c^{1/2}$. Recall now that since $r^2 = 1$, we have $r = \pm 1$. If $r = -1$, the above implies that all even coefficients in Eq. (5.12) are zero. Similarly, if $r = 1$ then all odd ones must be zero. Factoring this polynomial, we then find that

$$\Lambda(u) = \rho(\sin u \cos u)^{-p} \prod_{j=1}^{\ell} \sin(u - u_j),$$

where $\rho, u_1, \dots, u_j$ are constants yet to be determined and $\ell = 2p$ if $r = 1$ and $2p - 1$ if $r = -1$. To determine the constants $u_1, \dots, u_j$, one simply subs the above expression for $\Lambda(u)$ into the inversion identity Eq. (5.11) and does some simplifying algebra (a clever change of variables will help). Doing so, we find that

$$\Lambda(u) = \rho(\sin u \cos u)^{-p} \prod_{j=1}^{\ell} \sin(u + i\phi_j + \frac{1}{4}\gamma_j\pi),$$

where

$$\phi_j = \frac{1}{2} \log(\tan(\theta_j/2)), \quad \theta_j = \begin{cases} \pi(j - \frac{1}{2})/2p & \text{if } r = 1, \\ \pi j/2p & \text{if } r = -1, \end{cases}$$

and $\gamma_j \in \pm 1$. Note that we can now also solve for the constant ρ by substituting *this* expression for $\Lambda(u)$ into the inversion identity Eq. (5.11), so we can regard this constant as known. Finally, we wish to get $\Lambda(u)$ into a form that is easier to maximize and asymptotically analyze. In light of this, we consider the function

$$M(u) = \tau(-4)^p [(\sinh 2L)^{2p} + r(\sinh 2K)^{2p}] \prod_{j=1}^{2p} (\mu_j)^{\gamma_j}, \quad (5.13)$$

where

$$\mu_j = \frac{\cosh 2K \cosh 2L + c_j}{\exp(i\theta_j) \sinh 2K + \exp(-i\theta_j) \sinh 2L}, \quad c_j = (2 - 2 \cos 2\theta_j)^{1/2}$$

and $\tau = 1$ if $r = 1$ and $-i$ if $r = -1$. One can argue as per standard complex analytical procedure (namely, by showing that Λ^2 and M have the same poles with the same principal parts, that the difference $\Lambda^2 - M$ is bounded and hence constant, and that this constant is zero) to show that $\Lambda^2 = M$.

5.2.7 Calculation of the Free Energy

The benefit of the identity $\Lambda^2 = M$ is that it expresses the eigenvalue function Λ in a way that can be easily maximized and asymptotically analyzed, thereby allowing us to explicitly compute f . For instance, because

$$\mu_j \overline{\mu_j} = \frac{\cosh 2K \cosh 2L + c_j}{\cosh 2K \cosh 2L - c_j},$$

and it is not hard to see that $0 \leq c_j \leq \cosh 2K \cosh 2L$ for each j , we have that $|\mu_j| \geq 1$. It is immediate from this that $M = \Lambda^2$ is maximized by choosing $\gamma_j = 1$ for each j . Now recall that $Z_N \sim \Lambda_{\max}^m$, where $\Lambda_{\max}$ denotes the eigenvalue of VW of maximum modulus. Since each row in the lattice has $n = 2p$ sites, there are $N = 2mp$ sites in the entire lattice. For large p , then, we have

$$f \approx -kT \frac{\log \Lambda_{\max}^m}{2mp} = -kT \frac{\log \Lambda_{\max}}{2p}.$$

The miracle is that, once choosing $\gamma_j = 1$ for each j , the product of the denominators of the μ_j exactly cancels with the leading factors in Eq. (5.13), except for a factor of 4^p which can be distributed into the product to yield

$$\Lambda_{\max}^2 = \prod_{j=1}^{2p} 2(\cosh 2K \cosh 2L + c_j).$$

This identity holds for both $r = \pm 1$. Note, however, that we must in fact have $r = 1$, since the Perron-Frobenius theorem implies that the maximum eigenvalue of a matrix with all positive entries corresponds to an eigenvector with all positive entries (meaning that x has all positive entries and hence, since the matrix R has only non-negative entries, that the equation $Rx = rx$ can only be satisfied if $r = 1$). Consider, now, the function F defined by

$$F(\theta) = \log \left[2(\cosh 2K \cosh 2L + (2 - 2 \cos 2\theta)^{1/2}) \right].$$

Then, since we know r to be equal to 1 and hence that $\theta_j = \pi(j - \frac{1}{2})/2p$, we have

$$\log \Lambda_{\max} = \frac{1}{2} \sum_{j=1}^{2p} F(\theta_j).$$

The final step is to notice that the above summation is secretly a Riemann sum. In particular, it is clear that $\theta_j \in (0, \pi)$ for each j . By making cuts at $x = k\pi/2p$ for $k = 0, \dots, 2p$, one divides the interval $(0, \pi)$ into $2p$ subintervals of equal width $\pi/2p$, the j th one having θ_j in the center. We then have that

$$\sum_{j=1}^{2p} F(\theta_j) \frac{\pi}{2p} \rightarrow \int_0^\pi F(\theta) d\theta$$

as $p \rightarrow \infty$, so if we write $\log \Lambda_{\max}$ as

$$\log \Lambda_{\max} = \frac{p}{\pi} \sum_{j=1}^{2p} F(\theta_j) \frac{\pi}{2p}$$

then for large p we have that

$$f \approx -kT \frac{\frac{p}{\pi} \sum_{j=1}^{2p} F(\theta_j) \frac{\pi}{2p}}{2p} = -\frac{kT}{2\pi} \sum_{j=1}^{2p} F(\theta_j) \frac{\pi}{2p}.$$

In the limit $p \rightarrow \infty$, we therefore arrive at the following beautiful formula:

$$f = -\frac{kT}{2\pi} \int_0^\pi F(\theta) d\theta.$$

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